Master Plan
2002-2022

Lincoln Township
Clare County, Michigan

Adopted by the Lincoln Township Planning Commission at a regular meeting on: September 17, 2002

Prepared by:

Lincoln Township Planning Commission

and

Wade-Trim, Inc.
Planning Consultant

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Clare County, Michigan

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Resolution for Adoption

Township of Lincoln
Planning Commission
PC Resolution No. 0201

A Resolution For The Adoption of A Master Plan For The Township of Lincoln.

WHEREAS, the Township of Lincoln, Clare County, State of Michigan, Planning Commission, with the assistance of Wade-Trim, Inc., Planning Consultant, has developed a Master Plan under the authority of the Township Planning Act, Public Act 168 of 1959, as amended; and,

WHEREAS, the Township of Lincoln, State of Michigan, envisions the Master Plan to be used as a guide for the future development of the Township of Lincoln; and,

WHEREAS, a Public Hearing on the Master Plan was held on September 17, 2002, at 7:00 p.m. at the Lincoln Township Hall, 175 Lake George Avenue, Lake George, Michigan.

NOW, THEREFORE BE IT RESOLVED, the Township of Lincoln Planning Commission formally adopts the Master Plan.

Moved by Commissioner Osborn and supported by Commissioner Galloway as follows:

Voting Yes: 5

Voting No: 0

Absent: 0

I declare this resolution adopted:

Al Lambert

Al Lambert, Chair

Attest:

Peg Taylor

Peg Taylor, Secretary

Table of Contents

Subject	Page No.
1.0 Introduction	**1**
1.1 Authority to Plan	1
1.2 Purpose of the Plan	1
1.3 Public Participation Process	2
1.4 Plan Organization	2
1.5 Location and Regional Context	3
2.0 Executive Summary	**5**
2.1 Background Information	5
2.1.1 Population	5
2.1.2 Housing	5
2.1.3 Economy	5
2.1.4 Natural Resources	6
2.1.5 Existing Land Use	6
2.1.6 Transportation	6
2.2 Community Goals and Objectives	6
2.3 Future Land Use Plan	6
2.4 Plan Implementation	7
3.0 Socioeconomic Profile	**8**
3.1 Population Trends	8
3.2 Population Projections	9
3.3 Age Distribution	10
3.4 Racial Make Up	11
3.5 Household Size	12
3.6 Household Characteristics	12
3.7 Total Housing Stock	13
3.8 Home Ownership	14
3.9 Age of Structures	15
3.10 Housing Values and Rent	16
3.11 Income and Poverty	17
3.12 Educational Attainment	18
3.13 Employment	18
3.14 State Equalized Values	20
4.0 Natural Resources Assessment	**21**
4.1 Climate	21
4.2 Geology	22
4.3 Topography	22
4.4 Woodlands	22
4.5 Wetlands	23
4.6 Soil Associations	23
4.7 Sites of Environmental Contamination	26
4.8 Leaking Underground Storage Tank (LUST) Sites	26

Subject	Page No.
5.0 Existing Land Use Analysis	**30**
5.1 Survey Methodology	30
5.2 Land Use Analysis	31
6.0 Transportation Analysis	**36**
6.1 Transportation Network	36
6.2 Transportation Analysis	37
7.0 Community Goals and Objectives	**40**
7.1 Introduction	40
7.2 Community-Wide Goals	40
7.3 Residential Goal and Objectives	40
7.4 Commercial Goal and Objectives	41
7.5 Industrial Goal and Objectives	41
7.6 Transportation Goal and Objectives	41
7.7 Infrastructure Goal and Objectives	42
7.8 Park and Recreation Goal and Objectives	42
7.9 Natural Environment Goal and Objectives	42
8.0 Future Land Use Plan	**44**
8.1 Introduction	44
8.2 What is a Future Land Use Plan?	44
8.3 Basis for the Plan	44
8.4 Plan Recommendations	45
8.5 Buildout Analysis	49
8.6 Using the Plan	51
9.0 Plan Implementation Resources	**54**
9.1 Keys to Successful Implementation	55
9.1.1 Knowledgeable Local Officials	55
9.1.2 Public Support	55
9.1.3 Co-Development	55
9.2 Implementation Tools	55
9.2.1 Zoning Ordinance	55
9.2.2 Subdivision Regulations	56
9.2.3 Enforcement	56
9.2.4 Capital Improvements Program	56
9.2.5 Michigan Certified Community Recreation Plan	57
9.2.6 Downtown Development Authority (DDA)	57
9.2.7 Special Assessment Districts	57

Table of Contents
(Continued)

Subject	Page No.
9.3 Grant and Loan Programs	57
9.3.1 Community Facilities Program	58
9.3.2 Public Works and Economic Development Program	59
9.3.3 Rural Business Enterprise Grants	59
9.3.4 Water and Wastewater Disposal Loan and Grant Program	60
9.3.5 Community Development Block Grant (CDBG)	60
9.3.6 Land and Water Conservation Fund (LWCF)	61
9.3.7 Michigan Natural Resources Trust Fund	61
9.3.8 Transportation Economic Development Fund	62
9.3.9 Transportation Enhancement Program (TEA-21)	62
9.4 Other Funding Options	63
9.4.1 Revenue Bonds	63
9.4.2 General Obligation Bonds	63
9.4.3 Local Funds	63
9.4.4 Local Businesses	63
9.4.5 Local Foundations	63
9.5 Sources of Information	63
9.5.1 Michigan Townships Association	63
9.5.2 County Extension Services	63
9.5.3 County Planning Commission	64
9.5.4 Regional Planning Agency	64
9.5.5 Other Local Officials	64
9.5.6 Libraries	64
9.5.7 Internet	64

Tables

Table No.	Page No.
1. Population Trends, Lincoln Township and Surrounding Communities, 1970-2000	9
2. Population Projections, Lincoln Township and Clare County, 2010 and 2020	10
3. Age Distribution, Lincoln Township and Surrounding Communities, 2000	11
4. Household Characteristics, Lincoln Township and Surrounding Communities, 2000	13
5. Type of Housing Units, Lincoln Township and Surrounding Communities, 2000	13
6. Housing Occupancy Characteristics, Lincoln Township and Surrounding Communities, 2000	15
7. Age of Housing Structures, Lincoln Township and Surrounding Communities, 2000	16
8. Distribution of Housing Values and Rent, Lincoln Township and Surrounding Communities, 2000	17
9. Income and Poverty, Lincoln Township and Surrounding Communities, 1989-1999	18
10. Employment by Industry, Lincoln Township and Surrounding Communities, 2000	19
11. State Equalized Values (in $), Lincoln Township, 1997-2002	20
12. Summary of Soil Characteristics, Lincoln Township	25
13. Existing Land Use Acreage, Lincoln Township, 2002	30
14. Future Land Use Acreage, Lincoln Township	45

Table of Contents
(Continued)

Tables

Table No.		Page No.
15.	Buildout Analysis, Lincoln Township	50
16.	Plan Implementation Resources	54
17.	Grant Programs by Project Type	58

Figures

Figure No.		Page No.
1.	Population Trends, Lincoln Township, 1940-2000	8
2.	Change in Median Age, Lincoln Township, Clare County, and Michigan, 1990-2000	11
3.	Racial Make Up, Lincoln Township, 2000	12
4.	Persons Per Household Trends, Clare County, 1980-2000	12
5.	Housing Occupancy and Ownership, Lincoln Township, 2000	14
6.	High School Graduates and Bachelor's Degree Holders, Lincoln Township and Surrounding Communities, 2000	19

Maps

Map No.		Page No.
1.	Geographic Location	4
2.	Topography	27
3.	Environmental Resources	28
4.	Soils	29
5.	Existing Land Use	35
6.	Transportation Network	38
7.	Transportation Analysis	39
8.	Future Land Use	52
9.	Buildout Analysis	53

Photos

Name	Page No.
Lake George Community Entryway Sign	1
Shingle Lake	5
Bertha Lake	21
Lake George Commercial District	31
Great Lakes Gas Transmission Company	32
Bringold Road	36

1.0 Introduction

Planning is a way to guide change, minimize negative impacts, and maximize positive aspects of growth. Planning provides a means to iron-out issues before they develop into conflicts and provides an opportunity for coordination among all interested organizations within the Township. The Master Plan establishes a framework through which the Township can structure future activities and carry out its goals and responsibilities. The Master Plan also serves as a tool which can project public service demands, anticipate traffic flow, project natural resources and determine natural limitations which affect the location of development. A balance is needed to accommodate growth pressures resulting from urban encroachment while preserving the composition and ecological balance of the Township. This cannot be accomplished without a commitment by local officials to carry out the planning process.

1.1 Authority to Plan

The Lincoln Township Planning Commission has prepared this master land use plan under the authority of the Township Planning Act, Public Act 168 of 1959, as amended. Section 6.(1) of the Act states:

The Planning Commission shall make and adopt a basic plan as a guide for the development of unincorporated portions of the township. As a basis for the plan, the planning commission may do any of the following:

a. *Make inquiries, investigations, and surveys of all the resources of the township.*

b. *Assemble and analyze data and formulate plans for the proper conservation and uses of all resources, including a determination of the extent of probable future need for the most advantageous designation of lands having various use potentials and for services, facilities, and utilities required to equip those lands.*



Lake George Community Entryway Sign

1.2 Purpose of the Plan

Section 2 of the Township Planning Act, Public Act 168 of 1959 gives a summary of the purpose of a master plan:

The purpose of plans prepared pursuant to this act shall be to promote public health, safety and general welfare; to encourage the use of resources in accordance with their character and adaptability; to avoid the overcrowding of land by buildings or people; to lessen congestion on public roads and streets; to facilitate provision for a system of transportation, sewage disposal, safe and adequate water supply, recreation and other public improvements; and to consider the character of each township and its suitability

for particular uses judged in terms of such factors as the trend in land and population development.

Planning is a process involving the selection of policies relating to land use and development in a community. The creation of a Master Plan is the first step of this process. A Master Plan is a statement of the goals and objectives for the future physical development of a community. Comprehensive and long term in nature, the Master Plan examines all aspects of the community, whether they are physical, social, or economic. A Master Plan performs three important functions:

- Provides a general statement of the community's goals and a comprehensive vision of the future.
- Provides the statutory basis for the Zoning Ordinance, as required by the Township Zoning Act, Michigan Public Act 184 of 1943, as amended.
- Serves as the primary policy guide for local officials considering development proposals, land divisions, capital improvements, and other matters related to land use and development; thus, providing a stable and consistent basis for decision making.

Every community's Master Plan is unique, focusing on important issues and challenges specific to that community. This Master Plan is designed to highlight Lincoln Township issues and to identify solutions to meet local needs.

1.3 Public Participation Process

This Master Plan was formulated through a process of active participation of the citizens of Lincoln Township. The Planning Commission held three public meetings for the purpose of involving citizens and the Commission to review and comment on the Plan at its three strategic phases.

Attended by the Planning Commission, Township Board of Trustee Members, and concerned citizens, the meetings were held in the Township Hall on the following dates:

1. May 21, 2002, Review of Background Information
2. June 18, 2002, Develop a consensus on Community Goals and Objectives
3. July 16, 2002, Finalize the Future Land Use Plan
4. September 17, 2002, Public Hearing

The May 21, 2002 meeting also included an open-ended discussion among those present at the meeting regarding the strengths, weaknesses, and what are the most desired characteristics of Lincoln Township in 2022. The discussion provided a basis for initial formulation of goals and objectives for the Township.

Pursuant to the public hearing notification requirements of Section 125.329, Township Planning Act 168 of 1959, as amended, the Planning Commission on September 17, 2002, conducted a public hearing on the Plan. Pursuant to the comments received from those present at the meeting, the Planning Commission at the meeting adopted the Plan.

1.4 Plan Organization

The Master Plan report is comprised of ten sections. The first section of the Master Plan is the Introduction, which gives an overview of the Master Plan and Lincoln Township.

The second section is the Executive Summary, which serves as a summary of the major findings and conclusions of the Master Plan for quick reference purposes.

The next four sections, Socio-economic Profile, Natural Resources Assessment, Existing Land Use Analysis, and Transportation Analysis, form the background studies portion of the Master

Plan. These sections highlight important conditions, characteristics, and trends within Lincoln Township that have direct impacts on current and future land use.

Community Goals and Objectives form the seventh section of the Master Plan. The goals and objectives are formulated through citizen input and participation, and are presented to guide future development. The goals and objectives also take into account the key land use trends, population trends, natural resources, and other issues presented in the background information sections.

The eighth section of the Master Plan is a Residential Buildout Analysis. The buildout analysis is useful for projecting future households and population under various density options. It permits ready comparison of alternative development patterns or regulatory strategies.

The Future Land Use Plan forms the ninth section of the Master Plan. Based on the background information, goals and objectives, buildout analysis, and other information such as current market trends, the Future Land Use Plan specifies the extent and location of where various types of future development can be accommodated within the Township.

Plan Implementation is the last section of the Master Plan. It serves as a guide for implementing the recommendations of the Master Plan.

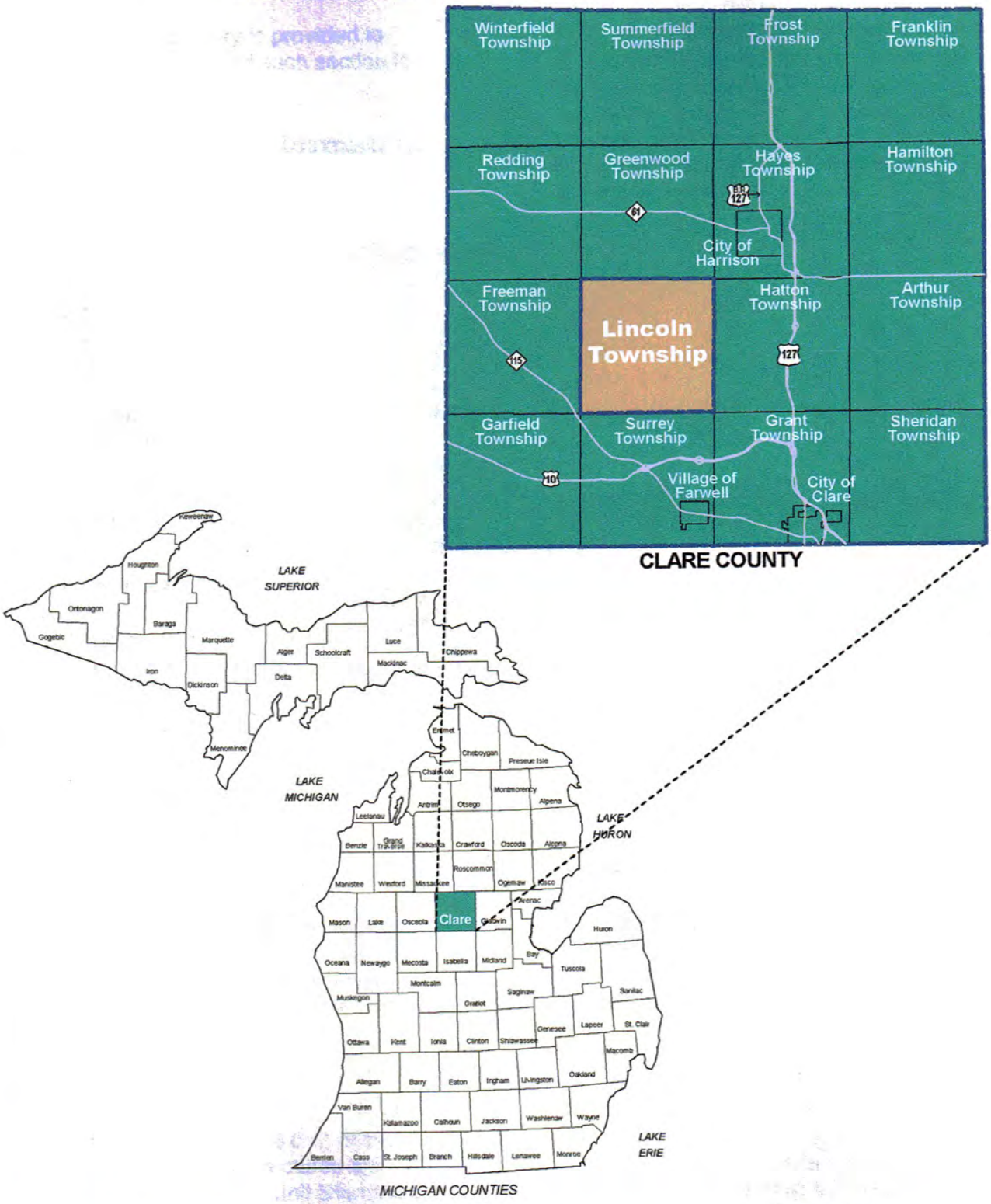
1.5 Location and Regional Context

Lincoln Township is located in the middle of Clare County in the region known as Central Michigan. Lincoln Township, as well as the rest of Clare County, is predominantly rural, known for its heavy forests, many lakes, and recreational opportunities. Because of these many assets, the region attracts thousands of tourists and seasonal residents throughout the year but particularly during the summer months.

Transportation access to Lincoln Township is provided by major state highways including U.S. 127, U.S. 10, M-61, and M-115, which are all located just outside the Township borders. The nearest major cities from Lincoln Township include Cadillac, Mount Pleasant, and the Tri-Cities of Bay City, Midland, and Saginaw, all within convenient driving distance.

Lincoln Township is made up of the geographic township T18N-R5W of Clare County. The unincorporated community of Lake George, located on the shores of Lake George, is located in the Township and serves as the focal point for commerce and services. Four Townships border Lincoln, including Greenwood to the north, Freeman to the west, Surrey to the south, and Hatton to the east. **Map 1** shows the geographic location of the Township.

Map 1
Geographic Location



Source: Michigan Resource Information System (MIRIS)

MASTER PLAN
LINCOLN TOWNSHIP
CLARE COUNTY, MICHIGAN

WADE-TRIM
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Bay City, Michigan 48706
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9/17/02

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2.0 Executive Summary

The Executive Summary is provided to highlight major findings of each section for quick reference purposes.

2.1 Background Information

2.1.1 Population

- Lincoln Township has experienced significant population growth in recent years. In 1970, the Township's population was 645 people. Throughout the next 30 years, the Township's population rose steadily until reaching 1,758 in the year 2000. Over the entire 30 year span, the Township grew by 172.6%.
- The Township's population is aging, which is evidenced by the change in median age from 43.7 in 1990 to 45.7 in 2000.
- The County persons per household average has decreased steadily over the years, declining from 2.72 in 1980 to 2.42 in 2000.
- The highest percentage of the Township's population are Married Couple Families at 57.8%. This percentage is similar to Clare County at 55.3%, but is higher than the State at 51.4%. The second largest household group in the Township is Single Persons Living Alone at 25.8%.

2.1.2 Housing

- A significant majority (78.0%) of the Township's housing structures are single-family homes (1 Unit Structures). The rest of the housing in the Township consists of Mobile Home or Trailer Units at 22.0%. Currently, Lincoln Township has no multi-family structures.
- Nearly two thirds (60.7%) of Lincoln Township's housing units are vacant. Of those vacant units, 93.7% are seasonally vacant. This means that the population of Lincoln Township swells significantly during peak seasons when seasonal residents arrive.



Shingle Lake

- Lincoln Township has a median housing value of $81,500, which is fairly high when compared to the surrounding Townships and Clare County. The State of Michigan's median housing value of $115,600 is significantly higher than Lincoln Township and all its neighbors.

2.1.3 Economy

- Lincoln Township's 1999 income levels (family, household, and per capita) are the highest among all surrounding Townships and Clare County. Only the State of Michigan has higher family, household, and per capita income levels. Lincoln Township's 1999 poverty percentage of 11.1% is slightly higher than the State at 10.5% but is lower than the County at 16.0%.

- In terms of Educational Attainment, Lincoln Township's high school graduate percentage of 78.0% is slightly higher than the County at 76.1% but is lower than the State at 83.4%. Lincoln Township's bachelor degree holder percentage of 7.8% is slightly lower than the County at 8.8% and is much lower than the State at 21.8%.
- Lincoln Township has a workforce well diversified in several industries. These industries include Manufacturing (23.8%); Educational, Health, and Social Services (17.5%); Wholesale and Retail Trade (14.5%); and Arts, Entertainment, and Recreation (13.6%).
- The Township's 2002 State Equalized Value is at $102,093,000, which represents a growth of 54.0% from its 1997 State Equalized Value.

2.1.4 Natural Resources

- Some minor topographical features such as hills and lakebeds are found in Lincoln Township, which may pose constraints to land development. Proper procedures for reducing soil erosion and containing water runoff should be utilized when development occurs in these areas.
- Heavy woodlands are a dominant feature in Lincoln Township. In total, woodlands (upland and lowland forests) cover well over three-quarters (82.5%) of the Township.
- Small concentrations of environmentally sensitive wetland areas are scattered throughout Lincoln Township. In total, wetlands cover 1,104 acres or 4.8% of the Township.

2.1.5 Existing Land Use

- Of the 22,967 total acres of land in the Township, the highest percentage (61.2%) is Vacant/Right-of-Way/Other.
- The second largest land use category is Single-Family Residential comprising 20.3% of the Township.
- A significant percentage (14.1%) of land in the Township is owned by gas companies. These properties are concentrated south of the Lake George area.

2.1.6 Transportation

- Lincoln Township's transportation network is somewhat fragmented, consisting of county primary roads that follow indirect routes with many 90 degree turns. The majority of the local roads in the Township are unpaved.

2.2 Community Goals and Objectives

- The Goals and Objectives for the Township are based upon an analysis of the background information in this Master Plan as well as a community survey conducted on May 21, 2002.
- The fundamental community-wide goal of the Township is to "preserve and protect the Townhsip's abundant natural resources such woodlands, lakes, and wetlands."
- Another important community-wide goal of the Township is to "capitalize on the Township's abundant resources for the purpose of creating an attractive community that meets the physical, social, and economic needs of its residents and businesses in an environmentally sensitive manner."

2.3 Future Land Use Plan

- In keeping with the Township's existing rural character, the highest percentage (51.6%) of the Township is classified as Rural Residential in the Future Land Use Plan. This category is designed to

preserve the area's existing rural character by protecting natural resources and discouraging large-scale developments.

- The second largest category is Single-Family Residential at 24.9%. The purpose of this district is to enhance the existing single-family residences, encourage new single-family development, and to promote orderly growth resulting in a cost-effective expansion of Township services.
- Other important recommendations presented in the Future Land Use Plan include the creation of a well-defined Central Business District, designation of strategic areas for new industrial developments, the identification of a potential public utility service area, and the identification of four Township Entryways.

2.4 Plan Implementation

The process of carrying out the goals and recommendations of this Master Plan can be challenging. The Plan Implementation section of the Master Plan acts as a general reference guide to be used by Township officials in order to help stimulate the implementation process. The section includes a series of keys to successful implementation, implementation tools, grant and loan program, other funding options and sources of information.

3.0 Socioeconomic Profile

When developing a Master Plan, it is important to perform background studies to identify the nature and character of the community's natural, human, and institutional environments. Socioeconomic characteristics are essential components in the development of any master plan and a continuing planning process. While an evaluation of the community's current situation provides insight into immediate needs and deficiencies, projections provide a basis for determining future land use requirements, public facility needs, and essential services.

By examining socioeconomic characteristics such as population, income, and employment base, a community can identify trends and opportunities that will influence future land use decisions and policy choices. Since certain socioeconomic analyses have an identifiable impact upon the future of a community, appropriate sections have been detailed to relate social trends to future economic considerations. Secondly, the socioeconomy of a Township does not function in a vacuum. Consequently, this analysis is benchmarked, when appropriate, to larger socioeconomic environments and trends.

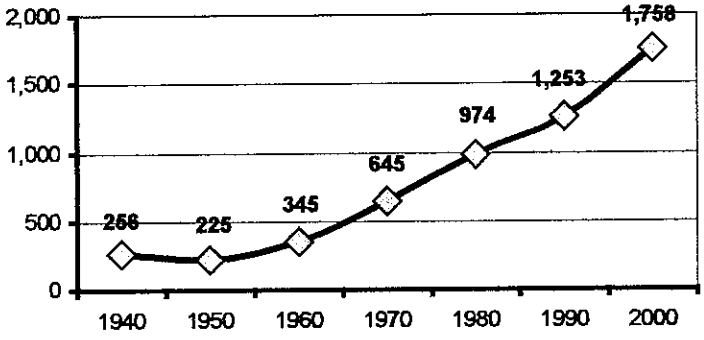
This socioeconomic profile uses the most recent 2000 U.S. Census data, as well as historical census data.

3.1 Population Trends

Population growth is the most important factor influencing land use decisions in any community. Simply put, if the population of a community is growing, there will be a need for more housing, commerce, industry, parks and recreation, public services and facilities, or roads.

Figure 1 shows the population trends for Lincoln Township since 1940. During the 20-year span between 1940 and 1960, the Township's population showed little change increasing by only 89 people. However, since 1960, the Township's population has been steadily increasing, growing by an average of 51.7% each decade.

Figure 1
Population Trends
Lincoln Township, 1940-2000



Source: 1940 through 2000 U.S. Census Reports

Table 1 compares Lincoln's population trends with the surrounding four Townships, Clare County, and the State of Michigan. Lincoln Township and its surrounding communities all experienced population growth between 1970 and 1980. The highest population growth rates occurred in Freeman Township (106.1%), Greenwood Township (79.3%) and Lincoln Township (51.0%). Between 1980 and 1990, every unit of government grew in population but at lower rates than during the previous decade. During the 80's, Freeman Township again experienced the highest growth of 40.3% with Lincoln Township following in second with a growth of 28.6%.

The most recent decade witnessed high population growth rates for Lincoln Township and its neighbors. Lincoln Township's growth of 40.3% was the third highest behind Freeman Township and Greenwood Township. Over the entire 30-year span, four Townships grew more than 100% including Lincoln Township at 172.6% while the County as a whole grew by 87.2%.

These high growth rates for Lincoln Township and surrounding communities are very impressive, especially when compared with Michigan's modest growth of 11.9% in the last 30 years.

Table 1
Population Trends
Lincoln Township and Surrounding Communities, 1970-2000

Lincoln Township	645	974	329	51.0	1,253	279	28.6	1,758	505	40.3	1,113	172.6
Freeman Township	212	437	225	106.1	613	176	40.3	1,118	505	82.4	906	427.4
Greenwood Township	362	649	287	79.3	718	69	10.6	1,059	341	47.5	697	192.5
Hatton Township	460	638	178	38.7	673	35	5.5	923	250	37.1	463	100.7
Surrey Township	2,338	3,101	763	32.6	3,221	120	3.9	3,555	334	10.4	1,217	52.1
Clare County	16,695	23,822	7,127	42.7	24,952	1,130	4.7	31,252	6,300	25.2	14,557	87.2
Michigan	8,881,826	9,262,044	380,218	4.3	9,295,297	33,253	0.4	9,938,444	643,147	6.9	1,056,618	11.9

Source: 1970, 1980, 1990, and 2000 U.S. Census.

3.2 Population Projections

Table 2 shows population projections for Lincoln Township and Clare County for the years 2010 and 2020. The data in the table has been compiled using various sources and projection techniques. Two outside sources have been used to estimate the population of Clare County in the future: Michigan Department of Management and Budget (MDMB) and Woods & Poole Economics, Incorporated. Three methods (arithmetic, constant percentage, and increasing county share) have been used to estimate the population of Lincoln Township for 2010 and 2020.

The two population projections for Clare County indicate continued population growth for 2010 and 2020. Using an average of the two projections, it is estimated that the population of Clare County will reach 31,870 by 2010, and 34,510 by 2020.

For Lincoln Township, all three population projection methods predict population growth for 2010 and 2020. Using an average of the three projection methods, it is estimated that the population of Lincoln Township will reach 2,190 by 2010 and 2,765 by 2020. **Table 2** also includes a population projection estimated by Wade-Trim using a 45% growth rate for the next two decades. Under this growth rate, the population of Lincoln Township will reach 2,549 by 2010 and 3,696 by 2020. This 45% predicted growth for the next two decades is based on many factors including:

- Current population movement from urban areas to rural areas with transportation access, desirable environmental features, and proximity to population centers.
- Lincoln Township taking a proactive role in attracting growth.

Table 2
Population Projections
Lincoln Township and Clare County, 2010 and 2020

Clare County	**16,695**	**23,822**	**24,952**	**31,252**	**--**	**--**
MDMB Projections (2)	--	--	--	--	33,100	36,900
Woods & Poole Projections (3)	--	--	--	--	30,640	32,120
Average	--	--	--	--	31,870	34,510
Lincoln Township	**645**	**974**	**1,253**	**1,758**	**--**	**--**
Arithmetic (4)	--	--	--	--	2,129	2,500
Constant Percentage (5)	--	--	--	--	2,461	3,445
Increasing County Share (6)	--	--	--	--	1,980	2,347
Average	--	--	--	--	2,190	2,764
Wade-Trim Estimate (7)	--	--	--	--	2,549	3,696

(1). Source: U.S. Census Bureau.
(2). Michigan Department of Management and Budget, Office of the State Demographer, 1996.
(3). Woods & Poole Incorporated, State Data Book, 1998.
(4). Population increases by the average increase in number between 1970 and 2000.
(5). Population increases by the average percentage increase between 1970 and 2000.
(6). Population is determined using the Township's 2000 share of the county population as the base, plus the average increase of the Township's share of the county population between 1970 and 2000.
(7). Wade-Trim estimated growth rate of 45% per decade.

3.3 Age Distribution

As humans progress through life, they pass through stages of life that generally correspond to their age levels. Age distribution analysis is used by demographers and policy makers to anticipate future changes in such areas as consumption, housing, medical care, education, and recreation. In this analysis, five life cycles are defined:

- Preschool (0-4 years)
- School (5-19 years)
- Family Formation (20-44 years)
- Empty Nest (45 to 64 years)
- Elderly (65 years or more)

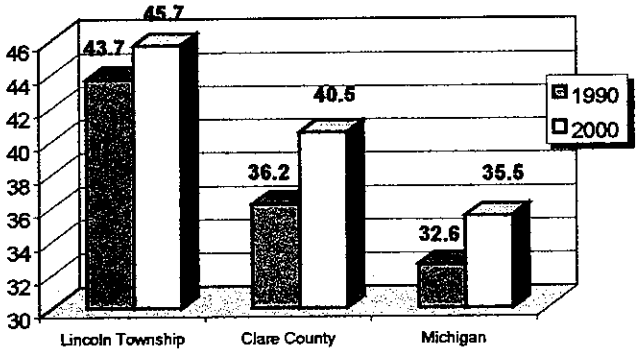
The "baby boomers" born between 1946 and 1964 are the single largest population group. As this population bulge moves through life, the median age of the community gradually increases.

These baby boomers desire larger homes, require health care, and spur construction of retail and manufacturing, offices, and transportation improvements. As they age, boomers will impact leisure and recreation markets, move into smaller homes, especially homes they can age in. Moving into the senior and elderly years, boomers may be moving to retirement communities or even to their second home bought many years earlier. In the near future as boomers swell the ranks of the seniors and elderly, demand for health care services and facilities will increase.

The "echo boom" children, born between 1978 and 1995, form a generation almost as large as the original baby boom. During the next 10 to 15 years, the echo boom generation will induce increased demand and need for education, recreation, and consumer goods. By the year 2010, the pace of decline among the family formation population will slow as the echo boom begin to start their own families, moving from rented apartments to single family starter homes.

Figure 2 shows the change in median age for Lincoln Township, Clare County, and Michigan. All three units of government experienced a rise in median age in the last decade. The greatest increase occurred in Clare County, where the median age rose 4.3 years while Lincoln Township experienced the smallest increase of 2.0 years. However, Lincoln Township is the oldest of the three units of government with a median age that is more than 10 years higher than the State.

Figure 2
Change in Median Age
Lincoln Township, Clare County, and Michigan, 1990-2000



Source: 1990 and 2000 U.S. Census

Table 3 compares the age distributions for the Township, County, State, and surrounding Townships in 2000. With the exception of Hatton Township and Michigan, the communities in the table all have similar age distributions with fairly high percentages in the empty nester (45-65 years) and elderly (65 years and over) age groups. Similarly, every unit of government except Hatton Township and Michigan has a very high median age. The oldest community is Freeman Township with a median age of 48.8 while Lincoln Township's median age of 45.7 years is the second oldest.

Table 3
Age Distribution
Lincoln Township and Surrounding Communities, 2000

	Lincoln		Freeman		Greenwood		Hatton		Township 4		Clare County		Michigan	
	#	%	#	%	#	%	#	%	#	%	#	%	#	%
0-4	97	5.5	44	3.9	50	4.7	56	6.1	183	5.1	1,815	5.8	672,005	6.8
5-19	291	16.6	209	18.7	198	18.7	254	27.5	707	19.9	6,582	21.1	2,212,060	22.3
20-44	465	26.5	253	22.6	283	26.7	271	29.4	1,066	30.0	9,216	29.5	3,604,383	36.3
45-64	556	31.6	369	33.0	340	32.1	235	25.5	925	26.0	8,241	26.4	2,230,978	22.4
65+	349	19.9	243	21.7	188	17.8	107	11.6	674	19.0	5,398	17.3	1,219,018	12.3
TOTALS	**1,758**	**100.0**	**1,118**	**100.0**	**1,059**	**100.0**	**923**	**100.0**	**3,555**	**100.0**	**31,252**	**100.0**	**9,938,444**	**100.0**
Median Age	45.7		48.8		44.8		35.4		41.8		40.5		35.5	

Source: 2000 U.S. Census.

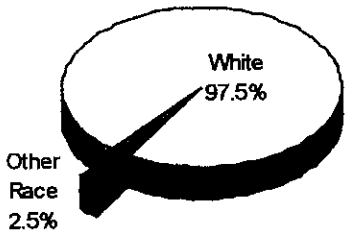
3.4 Racial Make Up

Another important characteristic of a community is its racial make-up. Knowing the racial make-up of a community helps to identify the diverse needs of its population. The census bureau separates race into six different categories: White, Black or African American, American Indian and Alaska Native, Asian, Native Hawaiian and Other Pacific Islander, and Some Other Race.

Even though the population of Lincoln Township has become slightly more diverse since 1990, the Township has remained

almost exclusively White. In 2000, 97.5% of the 1,758 persons in the community were White, declining from 98.7% in 1990. All other units of government in the area are similar to the Township, with high percentages of White persons including Clare County at 97.4%. Michigan has a much more diverse population with 80.2% White and 19.8% falling into the other five racial categories mentioned above. **Figure 3** shows the racial distribution of Lincoln Township's population in 2000.

Figure 3
Racial Make-Up
Lincoln Township, 2000



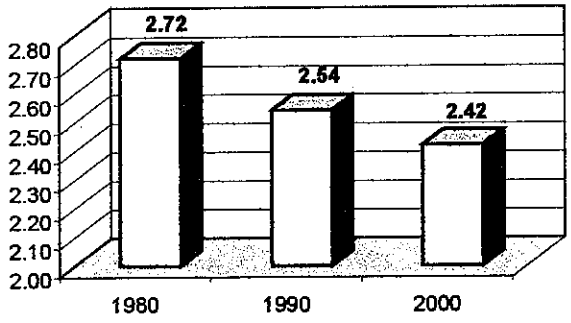
Source: 2000 U.S. Census

3.5 Household Size

The number of persons per household constitutes household size. Since the 1970's, the nationwide trend has been a decline in household size. This trend has occurred because of a number of reasons which include: declining number of children per family, higher divorce rates, growing number of elderly living alone, and the growing number of young people moving away from families to live on their own. Knowing whether the household size is increasing or decreasing is very important. If the household size of a community is decreasing, this means that new housing units might be necessary to accommodate for more citizens needing places to live. This can even be true if the overall population of a community is declining. In some municipalities, new housing units are being built to accommodate the demand for housing created by lower household sizes in spite of declining overall populations.

The persons per household trends for Clare County since 1980 are shown in **Figure 4**. The figure clearly shows how the number of persons per household has declined in Clare County in the past 20 years. In 1980, the Township had a persons per household number of 2.72, which declined to 2.54 by 1990, and continued to decline until reaching 2.42 in 2000.

Figure 4
Persons Per Household Trends
Clare County, 1980-2000



Source: 1980, 1990, and 2000 U.S. Census

3.6 Household Characteristics

This section examines households in terms of the relationships among the persons who share a housing unit. Some households are families, consisting of two or more persons related by blood, marriage, or adoption, while others are non-family households composed of persons living alone or with unrelated persons.

Household characteristics for the Township and other surrounding communities in 2000 are compared in **Table 4**. The highest percentage of households in Lincoln Township are Married Couple Families at 57.8%. This percentage is slightly higher than the county at 55.3% and much higher than the state at 51.4%. Other significant percentages in Lincoln Township include Single Persons Living Alone (25.8%) and Single Female Families (8.4%).

Table 4
Household Characteristics
Lincoln Township and Surrounding Communities, 2000

Married Couple Families	443	57.8	297	63.3	271	61.9	203	61.0	783	54.2	7,010	55.3	1,947,710	51.4
Single Male Families	31	4.0	19	4.1	21	4.8	22	6.6	53	3.7	548	4.3	154,187	4.1
Single Female Families	64	8.4	31	6.6	36	8.2	20	6.0	149	10.3	1,191	9.4	473,802	12.5
Single Persons Living Alone	198	25.8	94	20.0	89	20.3	67	20.1	391	27.1	3,330	26.2	993,607	26.2
Other Non-Families	30	3.9	28	6.0	21	4.8	21	6.3	68	4.7	607	4.8	216,355	5.7
TOTAL HOUSEHOLDS	**766**	**100.0**	**469**	**100.0**	**438**	**100.0**	**333**	**100.0**	**1,444**	**100.0**	**12,686**	**100.0**	**3,785,661**	**100.0**
Average Household Size	2.3		2.38		2.42		2.77		2.37		2.42		2.56	

Source: 2000 U.S. Census.

3.7 Total Housing Stock

Housing stock is the most basic measure of housing that refers to the type of housing units found in a community. The U.S. Census Bureau separates housing units into four different categories: 1 Unit Structures (single-family homes), Units in 2-4 Unit Structures (e.g. duplex units), Units in 5 or More Unit Structures (e.g. apartment units), and Mobile Home or Trailer Units. **Table 5** shows the distribution of housing units for the Township, surrounding Townships, and Clare County in 2000.

Table 5
Type of Housing Units
Lincoln Township and Surrounding Communities, 2000

1 unit structures	1,520	78.0	762	81.3	603	69.6	335	72.5	1,563	79.0	16,997	76.5
Units in 2-4 Unit Structures	0	0.0	0	0.0	2	0.2	2	0.4	22	1.1	280	1.3
Units in 5 or More Unit Structures	0	0.0	2	0.2	0	0.0	10	2.2	64	3.2	674	3.0
Mobile Home or Trailer Units	429	22.0	173	18.5	262	30.2	115	24.9	328	16.7	4,278	19.2
TOTALS	**1,949**	**100.0**	**937**	**100.0**	**867**	**100.0**	**462**	**100.0**	**1,977**	**100.0**	**22,229**	**100.0**

Source: 2000 U.S. Census.

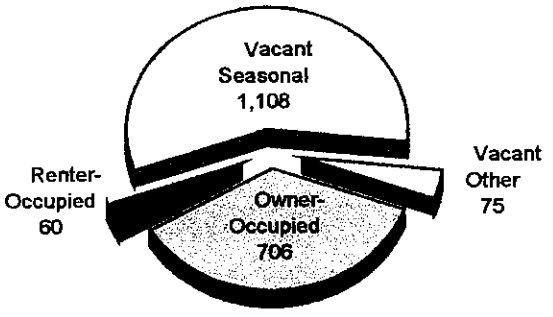
Lincoln Township has a high percentage of 1 Unit Structures (single-family homes) at 78.0%. This percentage is comparable to the surrounding townships and Clare County. The second highest percentage of housing structures in the Township are Mobile Home or Trailer Units at 22.0%. Lincoln Township currently has no multiple-unit structures.

3.8 Home Ownership

The home ownership rate is a very important aspect of a community. A high home ownership rate may mean that a community has stable and well kept neighborhoods. A higher percentage of rental housing might mean unstable neighborhoods with frequent resident changes as well as poorly kept properties.

The U.S. Census Bureau divides housing units into two categories: Occupied Housing and Vacant Housing. Within those categories, housing units are further divided into owner-occupied, renter-occupied, vacant seasonal, and vacant other. **Figure 5** shows the distribution of housing units according to the four categories as identified by the U.S. Census. The figure clearly shows that seasonal residents occupy the majority of the housing stock (1,108 units) in the Township. Owner-occupied housing units account for 706 units in Lincoln Township. Such a high number of seasonal housing units indicates that the population of Lincoln Township increases significantly during certain seasons of the year, especially during the summer.

Figure 5
Housing Occupancy and Ownership
Lincoln Township, 2000



Source: 2000 U.S. Census

Table 6 shows more detailed Census statistics concerning housing characteristics for Lincoln Township and the surrounding communities. Of all the units of government compared in the table, Lincoln Township has the lowest percentage of occupied housing at 39.3%. As discussed above, this low percentage is due to the large number of seasonally vacant housing units. In terms of owner verses renter occupancy, Lincoln Township is comparable to its neighbors with an ownership rate of 92.2%.

Table 6
Housing Occupancy Characteristics
Lincoln Township and Surrounding Communities, 2000

Occupied Housing	**766**	**39.3**	**469**	**50.3**	**438**	**50.9**	**333**	**71.5**	**1,444**	**72.6**	**12,686**	**57.1**
Owner-Occupied	706	92.2	435	92.8	402	91.8	292	87.7	1,211	83.9	10,437	82.3
Renter-Occupied	60	7.8	34	7.2	36	8.2	41	12.3	233	16.1	2,249	17.7
Vacant Units	**1,183**	**60.7**	**464**	**49.7**	**423**	**49.1**	**133**	**28.5**	**545**	**27.4**	**9,543**	**42.9**
Vacant Seasonal	1,108	93.7	440	94.8	393	92.9	105	78.9	424	77.8	8,583	89.9
Vacant Other	75	6.3	24	5.2	30	7.1	28	21.1	121	22.2	960	10.1
TOTALS	**1,949**	**100.0**	**933**	**100.0**	**861**	**100.0**	**466**	**100.0**	**1,989**	**100.0**	**22,229**	**100.0**

Source: 2000 U.S. Census.

3.9 Age of Structures

Analyzing the age of housing units is a way to measure the physical quality of the total housing stock of a community. The U.S. Census Bureau divides housing units into four categories according to the year the structure was built: 1939 or earlier, 1940 to 1959, 1960 to 1979, and 1980 to 1900. These groupings are helpful in determining the economic viability of housing structures. Any housing unit classified into the oldest two housing age categories could be in need of rehabilitation or repairs. However, some of the older housing of a community might be very well built, as well as desirable because of historical or architectural value, while at the same time newer housing might not be of good quality.

Table 7 shows the percentage of housing units in each category for Lincoln Township and the surrounding communities. Both Lincoln Township and Freeman Township have the lowest percentage of units built earlier than 1939 at 4.2%. In comparison, 16.9% of Michigan's housing units were built earlier than 1939. Lincoln Township's low older housing percentage of 4.2% may be an indicator that the majority of the housing in the Township is not in need of rehabilitation or repairs. In terms of new housing (built between 1980 and 1999), Lincoln Township's percentage of 33.2% is comparable to the surrounding townships and county.

Table 7
Age of Housing Structures
Lincoln Township and Surrounding Communities, 2000

Lincoln Township	4.2%	24.9%	37.7%	33.2	100.0%
Freeman Township	4.2%	6.6%	31.3%	57.9	100.0%
Greenwood Township	4.9%	8.8%	46.6%	39.7	100.0%
Hatton Township	11.9%	13.9%	34.4%	39.8	100.0%
Surrey Township	6.5%	18.2%	43.0%	32.3	100.0%
Clare County	6.8%	19.8%	44.1%	29.3	100.0%
Michigan	16.9%	26.5%	31.3%	25.3	100.0%

Source: 2000 U.S. Census

3.10 Housing Values and Rent

Analyzing housing values and rent could be the best way to determine both the quality and affordability of housing. It is of crucial importance that a community maintains both quality and affordable housing.

The distribution of housing and rental values for the Township, surrounding Townships, Clare County, and the State of Michigan in 2000 are shown in **Table 8**. Also shown in **Table 8** are the median housing and rental values for each unit of government.

The majority (48.0%) of Lincoln Township's housing units are valued between $50,000 and $99,000. Every other unit of government also has the majority of their housing units falling into this category. In terms of housing units valued less than $50,000, Lincoln Township's percentage of 23.5% is lower than Clare County at 31.3% but higher Than Michigan at 9.9%.

The median housing values for the Townships and Clare County are similar falling within a range of about $60,000 to $80,000. In comparison, Michigan has a much higher median housing value of $115,600. The median rental values for the Townships and Clare County are also low, falling within a range of about $375 to $525 as compared to Michigan's median rental value of $546.

Table 8
Distribution of Housing Values and Rent
Lincoln Township and Surrounding Communities, 2000

VALUE: Specified owner-occupied housing units*	**485**	**100.0**	**290**	**100.0**	**163**	**100.0**	**110**	**100.0**	**780**	**100.0**	**6,532**	**100.0**	**2,269,175**	**100.0**
Less than $50,000	114	23.5	34	11.7	56	34.4	18	16.4	227	29.2	2,043	31.1	224,603	9.9
$50,000-$99,999	233	48.0	158	54.5	78	47.9	64	58.1	381	48.8	2,967	45.4	711,648	31.4
$100,000-$149,999	94	19.4	78	26.9	16	9.8	16	14.5	99	12.7	883	13.5	603,454	26.6
$150-$199,999	34	7.0	14	4.8	10	6.1	6	5.5	65	8.3	444	6.8	339,716	15.0
Over $200,000	10	2.1	6	2.1	3	1.8	6	5.5	8	1.0	195	3.0	389,754	17.1
Median Value	81,500		83,700		65,000		75,000		72,200		70,500		115,600	
RENT: Specified renter-occupied housing units**	**52**	**100.0**	**40**	**100.0**	**32**	**100.0**	**28**	**100.0**	**244**	**100.0**	**2,079**	**100.0**	**976,313**	**100.0**
Less than $300	4	7.7	3	7.5	2	6.3	2	7.1	51	20.9	523	25.2	105,874	10.8
$300-$499	18	34.6	12	30.0	24	75.0	14	50.0	103	42.2	914	44.0	275,832	28.3
$500-$749	17	32.7	17	42.5	2	6.2	8	28.6	68	27.9	387	18.6	373,820	38.3
Over $750	4	7.7	2	5.0	0	0.0	0	0.0	8	3.3	67	3.2	178,021	18.2
No Cash Rent	9	17.3	6	15.0	4	12.5	4	14.3	14	5.7	188	9.0	42,766	4.4
Median Rent	497		517		436		467		429		397		546	

* Specified housing units include only one-family houses on less than ten acres without a commercial establishment or medical office on the property.
** Contract rent is the monthly rent agreed to, or contracted for, regardless of any furnishing, utilities, or services that may be included.
Source: 2000 U.S. Census.

3.11 Income and Poverty

Studying income and poverty levels is a good way to measure the relative economic health of a community. Three measures of income (median household, median family, and per capita) are illustrated in **Table 9** for the Township, County, State, and surrounding Townships. Household income is a measure of the total incomes of the persons living in a single household. Family income is a measure of the total incomes of a family unit. Because families often have two incomes, and do not include single persons living alone, median family incomes are typically higher than median household incomes. Per capita income is a measure of the incomes of every citizen of an area, including children. Because per capita income is based on the average of all individuals, they are much lower than family or household incomes.

In order to account for inflation between 1989 and 1999, the family, household, and per capita income numbers from 1989 have been altered. According to the Consumer Price Index, $1.00 in 1989 is equal to the value of $1.37 in 1999. Therefore, the actual incomes from 1989 were multiplied by 1.37 to come up with the figures shown in the table below.

Table 9
Income and Poverty
Lincoln Township and Surrounding Communities, 1989-1999

Lincoln Township	$25,385	$38,304	$21,971	$32,279	$12,123	$18,146	22.2%	11.1%
Freeman Township	$24,464	$33,333	$21,161	$30,398	$10,887	$14,624	31.8%	13.0%
Greenwood Township	$34,060	$35,703	$23,594	$32,228	$9,648	$17,283	17.6%	16.0%
Hatton Township	$39,559	$33,229	$29,186	$30,461	$14,923	$15,848	23.8%	11.3%
Surrey Township	$31,879	$32,475	$24,604	$28,350	$10,745	$17,027	19.2%	11.0%
Clare County	$28,323	$33,934	$23,513	$28,845	$12,583	$15,922	23.7%	16.0%
Michigan	$50,213	$53,457	$42,497	$44,667	$19,391	$22,168	10.2%	10.5%

*1989 dollars are multiplied by 1.37 to account for inflation when comparing to 1999 dollars.
Source: 1990 and 2000 U.S. Census.

Table 9 shows the family, household, and per capita income levels for the Township and surrounding units of government for 1989 and 1999. Over the past decade, Lincoln Township witnessed significant increases in incomes. In 1989, Lincoln Township's incomes were somewhat low in comparison to the surrounding Townships and County but by 1999, the Township's incomes grew to be the highest of all the surrounding Townships and County in all three income categories.

Table 9 also gives the percentages of families who were found to be below the poverty level. The percentage for Lincoln Township declined significantly from 22.2% in 1989 to 11.1% in 1999. This poverty percentage is one of the lowest of all the units of government compared in the table.

3.12 Educational Attainment

Educational attainment is another important factor in analyzing the capabilities of the local work force and the economic vitality of the community. The educational attainment of the citizens of a community plays a major role in determining what types of employment industries are suitable or necessary.

The U.S. Census Bureau reports on the percentage of citizens in a community who graduated from high school as well as those who went on after high school to complete bachelor's degrees. **Figure 6** illustrates the educational attainment levels for Lincoln Township as well as the surrounding Townships, Clare County, and Michigan in 2000. Lincoln Township's high school graduate percentage of 78.0% is the second highest of all the surrounding Townships but is slightly low when compared to Michigan at 83.4%. Similarly, Lincoln Township's bachelor's degree holder percentage of 7.8% is the second highest of all the surrounding Townships but is much lower than Michigan at 21.8%.

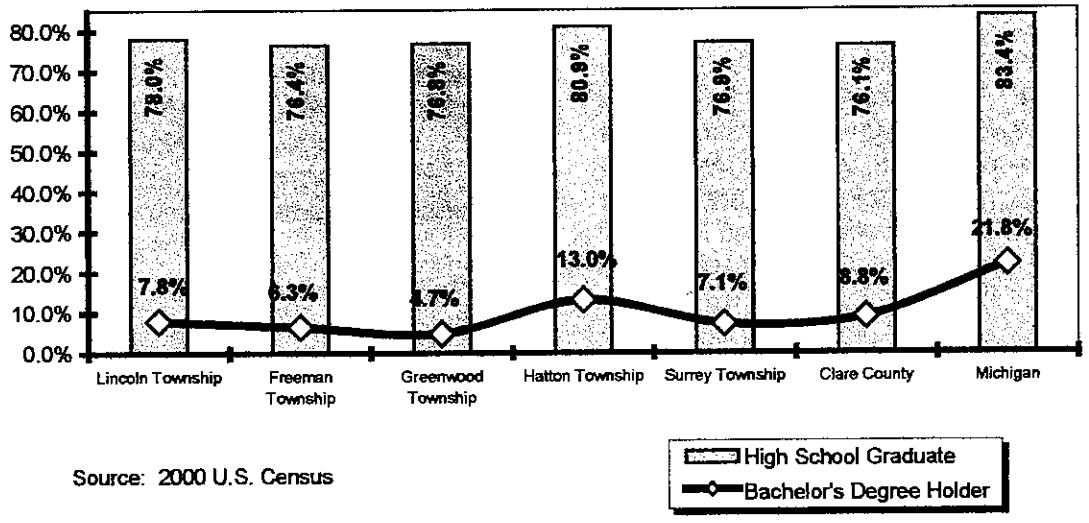
3.13 Employment

The U.S. Census Bureau presents employment data in terms of employment by industry. Employment by industry is a good way of analyzing what type of industries employ the citizens of Lincoln Township.

Table 10 shows the Employment by Industry for Lincoln Township, Clare County, and surrounding Townships in 2000. As can be seen by the table, Lincoln Township has four primary industries, which

employ a high percentage of its citizens. The four industries are: Manufacturing (23.8%), Educational, Health and Social Services (17.5%), Wholesale and Retail Trade (14.5%), and Arts, Entertainment, and Recreation (13.6%). Lincoln Township's employment distributions are very similar to the surrounding Townships and County.

Figure 6
High School Graduates and Bachelor's Degree Holders
Lincoln Township and Surrounding Communities, 2000



Source: 2000 U.S. Census

Table 10
Employment by Industry
Lincoln Township and Surrounding Communities, 2000

Industry	Lincoln Township		Freeman Township		Greenwood Township		Hatton Township		Surrey Township		Clare County	
	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent
Agriculture, Forestry, Fishing and Hunting, and Mining	20	3.3	10	2.8	13	3.2	10	2.7	29	2.0	400	3.4
Construction	55	9.2	24	6.8	33	8.2	62	16.5	78	5.5	1,049	9.0
Manufacturing	143	23.8	96	27.0	78	19.3	37	9.9	313	22.0	2,047	17.5
Wholesale and Retail Trade	87	14.5	57	16.1	67	16.6	59	15.7	265	18.6	1,898	16.3
Transportation and Warehousing, and Utilities	15	2.5	14	3.9	24	5.9	13	3.5	59	4.1	491	4.2
Information	3	0.5	8	2.3	4	1.0	4	1.1	13	0.9	130	1.1
Finance, Insurance, and Real Estate	31	5.2	20	5.6	22	5.4	16	4.3	34	2.4	472	4.0
Professional, Administrative, and Management	11	1.8	19	5.4	17	4.2	18	4.8	75	5.3	495	4.2
Educational, Health, and Social Services	105	17.5	52	14.6	63	15.6	85	22.7	275	19.3	2,332	20.0
Arts, Entertainment, and Recreation	82	13.6	32	9.0	39	9.7	35	9.3	181	12.7	1,360	11.6
Public Administration and Other Services	49	8.2	23	6.5	44	10.9	36	9.6	101	7.1	1,001	8.6
TOTALS	**601**	**100.0**	**355**	**100.0**	**404**	**100.0**	**375**	**100.0**	**1,423**	**100.0**	**11,675**	**100.0**

Source: 2000 U.S. Census

3.14 State Equalized Values

State Equalized Value (SEV) provides an important overview of a community's wealth. By law the SEV, which constitutes a community's tax base, is equal to approximately one-half of the true market value of real property and certain taxable personal properties.

Lincoln Township's state equalized values for the years 1997 through 2002 are shown in **Table 11**. The equalized values for each year are divided into four real property categories (agricultural, commercial, industrial, and residential) and one category for personal property.

By examining the table, it is clear that the state equalized values for Lincoln Township have been on the rise in recent years. The total state equalized value of the Township grew by 54.0% between 1997 and 2002 and is now at over 100 million dollars.

Within the four real property categories, commercial state equalized values experienced the highest growth of 117.9%. Agricultural and residential state equalized values also grew by high percentages at 91.9% and 91.0%, respectively. In terms of sheer numbers, residential property is by far the largest category of real property valued at more than 72 million dollars out of the total real property value of approximately 75 million dollars.

Table 11
State Equalized Values (in $)
Lincoln Township, 1997-2002

Real Property	**39,752,200**	**44,504,500**	**49,239,500**	**56,708,800**	**65,538,300**	**74,840,300**	**88.3**
Agricultural	7,400	8,700	10,100	10,100	11,800	14,200	91.9
Commercial	435,200	482,100	488,400	531,600	693,500	948,500	117.9
Industrial	1,498,300	1,606,200	1,702,900	1,696,300	1,827,300	1,665,100	11.1
Residential	37,811,300	42,407,500	47,128,100	54,470,800	63,005,700	72,212,500	91.0
Personal Property	**26,541,400**	**22,195,000**	**27,389,700**	**28,348,000**	**28,217,200**	**27,252,700**	**2.7**
Totals	**66,293,600**	**66,699,500**	**76,719,200**	**85,056,800**	**93,755,500**	**102,093,000**	**54.0**

Source: Lincoln Township Assessor.

4.0 Natural Resources Assessment

The natural environment plays a major role in land development. The natural environment can significantly impact development such as a steep slope prohibiting the construction of any structure. Conversely, the natural environment can be effected by land development. An example would be the increased erosion potential caused by clearing vegetation. Thus, when preparing a Future Land Use Plan, it is important to examine the natural environment in order to determine where development is best suited, and where it should be discouraged.

In any environmentally sensitive area within a community, development should be prevented. Environmentally sensitive areas are lands whose destruction or disturbance will affect the life of a community by either:

1. Creating hazards such as flooding or slope erosion.
2. Destroying important public resources such as groundwater supplies and surface water bodies.
3. Wasting productive lands and non-renewable resources such as prime farmland.

Each of these effects is detrimental to the general welfare of a community, resulting in social and economic loss.

The purpose of this section is twofold. First, the goal is to identify areas in the Township that are most suited for development. The focus is on areas that will minimize development costs and provide amenities without adversely impacting the existing natural systems. The second goal is to identify land that should be conserved in its natural state and is most suitable for conservation, open space or recreation purposes.

Climate, geology, topography, woodlands, wetlands, and soil conditions are among the most important natural features impacting land use in Lincoln Township. Descriptions of these features follow.

4.1 Climate

The climate of Clare County is seasonal, as the region experiences considerable changes in temperatures and precipitation throughout the year. The temperature range in January for Clare County is between 10 to 28 degrees Fahrenheit, in July between 56 to 82 degrees Fahrenheit. The average number of days below zero degrees Fahrenheit is 17, while the average number of days above 90 degrees Fahrenheit is 10. The average growing season in Clare County lasts 126 days. In terms of annual precipitation, Clare County averages 31 inches of rainfall and 52 inches of snowfall per year.



Bertha Lake

4.2 Geology

The geology of Lincoln Township, as well as the entire Lower Peninsula of Michigan, is described in terms of surface geology or quaternary geology (materials deposited by continental glaciers) and bedrock geology (sedimentary rocks underlying the glacial deposits).

The quaternary geology of the Township developed 10,000 to 12,000 years ago through continental glacial activity. As the glaciers melted and retreated from the landscape, large amounts of sand, gravel, clay, and loam were deposited. Massive glacial lakes formed at the front of the retreating glaciers. Clare County was among those submerged in glacial water.

The melting glacial water was laden with fine soil particles, which eventually settled to the bottom, creating clay and loam soils. The glacial melt water streams also deposited fine sands into the shallow glacial lakes. The sand channels are several miles wide in places, but the sand in them is generally only five to ten feet thick.

The sand deposits were further altered by wave action from these glacial "Great Lakes," creating small sand dunes and low beaches across the landscape as the water levels declined and the lakes retreated to their current area of coverage.

The sub-surface geology of Lincoln Township is sedimentary bedrock that was laid down during the Pennsylvanian ages of the Paleozoic Era. Bedrock is covered by glacial deposits and, generally, depending upon the thickness of the glacial deposits, are located at depths from 40 to 300 feet below the surface. The bedrock was formed from ancient seas, which covered the area some 250 to 600 million years ago. The shallow marine seas deposited layers of silt, clay, sediments, marine animals, plants, coral, and other calcareous materials. These deposits formed sandstone, shale, coal, and limestone bedrock.

4.3 Topography

The overall topography of Lincoln Township can be characterized as flat or gently rolling, but there are several areas of the Township where steep slopes can be found. Within the Township, elevations range from a low of about 980 feet above sea level to a high of approximately 1,310 feet above sea level, for a total elevation change of about 330 feet. **Map 2** shows the topography of Lincoln Township using contour lines on top of a color digital elevation model. The digital elevation model for Lincoln Township shows elevation change by a graduated color scheme where orange and reds are higher elevations and greens are lower elevations.

As can be seen on the map, the lower elevations of the Township are found in the southwest and southeast portions of the Township while the higher elevations are found in the central and northeast portions of the Township. The highest elevation of over 1,310 feet is found in Section 9 of the Township. The lowest elevation of about 980 feet is found in the southeast corner of the Township.

Map 2 also shows the locations in the Township where slopes are greater than 10%. There are several areas of the Township where concentrations of steep slopes are found. Urban development in steep slope areas such as those indicated on the map is strongly discouraged due to environmental concerns such as erosion.

Aside from the steeper hills, the gently sloping hills or flat areas that characterize the majority of the Township pose few constraints to land development.

4.4 Woodlands

Woodlands information for Lincoln Township is derived from the Michigan Resources Information System (MIRIS) land use cover data provided by the Michigan Department of Natural Resources. The Michigan Department of Natural Resources

breaks up woodlands into two categories: upland forests and lowland forests. Upland forests include mostly central hardwood (oak) trees such as red oak, white oak, sugar maple, red maple, black cherry, beech, basswood, and ash. Tree species in the lowland forest include silver maple, green ash, aspen, cottonwood, and elm. **Map 3** shows the general locations of upland and lowland forests in Lincoln Township.

In total, woodlands cover 82.5% of the Township. The majority of forests in Lincoln Township are classified as upland forests (18,009 acres or 78.4% of the Township). Lowland Forests comprise 935 acres or 4.1% of the Township. As shown on **Map 3**, upland forests are found throughout all areas of the Township, while lowland forests are primarily concentrated in the western half of the Township.

Because of many benefits associated with wooded areas, woodlands should be seen as a real asset to the Township. For human inhabitants, forested areas offer scenic contrasts within the landscape and provide recreational opportunities such as hiking and nature enjoyment. In general, woodlands improve the environmental quality of the whole community by reducing pollution through absorption, reducing the chances of flooding through greater rainwater infiltration, stabilizing and enriching soils, moderating the effects of wind and temperature, and providing habitats for wildlife.

4.5 Wetlands

Wetlands are an important, though commonly overlooked, natural resource which provide both aesthetic and functional benefits. Through the years, over 70% of Michigan's wetlands have been destroyed by development and agricultural activities. In order to protect wetlands, the State of Michigan enacted Part 303, Wetlands Protection, of the Natural Resources and Environmental Protection Act, Public Act 451 of 1994. The wetland act authorizes the Michigan Department of Environmental Quality (MDEQ) to preserve certain wetland areas. The MDEQ may require permits before altering regulated wetlands and may prohibit development in some locations.

The MDEQ defines wetlands as "land characterized by the presence of water at a frequency and duration sufficient to support wetland vegetation or aquatic life and which is any of the following:

- Contiguous to the Great Lakes, an inland lake or pond, or a river or stream.
- More than five acres in size.
- Determined by the MDEQ to be essential to the preservation of the natural resources of the state from pollution, impairment, or destruction."

Any wetlands in the Township not meeting the criteria for wetlands as defined by the MDEQ can be protected by local control techniques. Such techniques include a local wetland ordinance, policies in this Master Plan directing incompatible land uses away from wetlands, and specific wetlands provisions in the Zoning Ordinance.

As shown on **Map 3**, Lincoln Township has several small concentrations of wetlands. In total, wetlands cover 1,104 acres or 4.8% of the Township. The majority of these wetlands are found in the western half of the Township.

4.6 Soil Associations

Soil characteristics help define the land's capacity to support certain types of land uses. Soils most suitable for development purposes are well drained and are not subject to a high water table. Adequate drainage is important for minimizing storm water impacts and the efficient operation of septic drain fields. Adequate depth to the water table is necessary to prevent groundwater contamination from septic

systems. A high water table also limits the construction of basements. Though civil engineering techniques can be employed to improve drainage and maintain adequate separation from the water table, such techniques are expensive to construct and maintain.

According to the General Soils Map of Clare County provided by the Soils Conservation Service, there are six soil associations found in Lincoln Township. The general locations of these associations are shown on **Map 4**. The map is not designed for site specific applications. Rather, it can be used to compare land suitability for large areas.

Each soil association is composed of several soil series. Each series making up one association may occur in another, but in a different pattern and/or combination. A description of the six soil associations within Lincoln Township follows. Also included is a table summarizing the characteristics of each soil association as they relate to the potential for land use and development (**Table 12**).

Montcalm-Menominee-Nester Association

Soils of this association can be found in two locations near the eastern and southern edges of the Township. These soils, which are nearly level to rolling and well or moderately well drained with sandy and loamy subsoil, compose approximately 7% of the Township. This soil association has good potential for farming and grazing and woodlands habitat areas. The potential for building site development is good in the flatter areas where this soil association is found.

Menominee – Iosco – Kawkawlin Association

A limited amount of this soil association is found near the intersection of Browns Road and Hickory Road in the southern portion of the Township. In total, the association comprises approximately 2% of the Township. Soils in this association are nearly level to gently rolling with well drained to somewhat poorly drained soils that have sandy and loamy subsoil. Common land uses found on the soils of this association include grazing lands and woodlands, with some areas of cultivation. Excessive wetness, susceptibility to soil blowing, and a high content of clay are the major problems for most uses.

Graycalm-Montcalm Association

This soil association is composed of nearly level to steep, somewhat excessively drained and well drained soils that have sandy and loamy subsoil. Found throughout the Township, this association composes the highest percentage of the Township at approximately 80%. The soils of this association commonly support woodlands, but may be cultivated or pastured. Susceptibility to soil blowing and erosion and doughtiness are the major problems for urban development.

Grayling Association

This soil association is composed of nearly level to steep, excessively drained soils that have sandy subsoil. One small area containing soils of this association is found in the eastern half of the Township. In total, the association comprises approximately 2% of the Township. Because of extreme doughtiness, the soils pose poor potential for farming and grazing. Woodland is generally the best use for soils in this association because of the deep rooting capacity of the trees. The potential for urban development is good in the flatter areas and fair to poor in the hilly to steep areas. This soil association is a good source of sand.

Rubicon-Croswell-Au Gres Association

Soils of this association are found in the western half of the Township surrounding Hemlock Lake, Perch Lake, and Silver Lake. In total, this association comprises approximately 5% of the Township. The

soils of this association are used mainly for woodland, but some areas are pastured. Susceptibility to soil blowing and droughtiness are the major problems for most uses. Other problems are a seasonal high water table in the Croswell and Au Gres soils, the slope of the Rubicon soils, and erosion on the Rubicon soils. Croswell and Au Gres soils have fair to poor potential for urban development because they are wet, while Rubicon soils have good potential for urban development in the flatter areas.

Lupton-Markey Association

Found surrounding the creek and riverbeds in the western half of the Township, this association is comprised of nearly level and very poorly drained soils with mucky subsoil. In total, the association covers approximately 4% of the Township. This association is used mainly for woodland, but some areas are in marsh grass or are covered with brush. Excessive wetness and instability are the major problems for most uses. Also, the soils are commonly flooded for long periods. Most areas are undrained because drainage outlets are very difficult to locate.

Table 12
Summary of Soil Characteristics
Lincoln Township

Soil Association	Approximate Percentage of Township	Characteristics
Montcalm-Menominee-Nester Association	7	Used for farming, grazing, and woodlands. Good urban development potential in flat areas.
Menominee-Iosco-Kawkawlin Association	2	Used mainly for grazing lands and woodlands with some cultivation. Major limitations to development include wetness, soil blowing, and high clay content.
Graycalm-Montcalm Association	80	These soils commonly support woodlands, but may be cultivated or pastured. Erosion and droughtiness are the major limitations to urban development.
Grayling Association	2	Best suited for woodlands. The potential for urban development is good in the flatter areas.
Rubicon-Croswell-AuGres Association	5	Used mainly for woodland. Fair to poor potential for urban development on Croswell and AuGres, good potential on Rubicon.
Lupton-Markey Association	4	Used mainly for woodland. Excessive wetness and instability are the major limitations to urban development.

4.7 Sites of Environmental Contamination

The Michigan Department of Environmental Quality (MDEQ) publishes a list of contaminated sites throughout Michigan. Contaminated sites are sites with any of the following characteristics:

- Sites where a hazardous substance has been released, deposited, or disposed of.
- Oil and gas contamination sites.
- Hazardous waste management sites.
- Solid waste management sites.

As shown on **Map 4**, there are four sites of environmental contamination within Lincoln Township. The four sites are also listed below.

1. Residential Wells Lake George, 162 Lake George Street.
2. Great Lakes Gas Transmission Co., Compressor Station #12.
3. Primark Property, Hemlock and Ashard Roads.
4. Oil & Gas Contamination Site, Near Hickory Road.

4.8 Leaking Underground Storage Tank (LUST) Sites

The Michigan Department of Environmental Quality (MDEQ) also publishes a list of Leaking Underground Storage Tank (LUST) sites throughout Michigan. A LUST site has the following characteristics:

- Sites where a release has occurred from an underground storage tank system.
- Sites where chemicals from an underground storage tank have been detected in the groundwater, surface water, or subsurface soils.

There are currently three LUST sites in Lincoln Township as shown on **Map 4** and listed below.

1. 490 Arthur Road
2. Lincoln Compressor Station, 3991 South Hickory Road
3. Silver Lake Party Store, 108 West North Street

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

LILLY LAKE

SHINGLE LAKE

LAKE GEORGE

BUNGO LAKE

BERTHA LAKE

HEMLOCK LAKE

KILBORNE LAKE

PERCH LAKE

SILVER LAKE

HALFMOON LAKE

MONROE RD.

ARTHUR RD.

MANNSIDING RD.

CEDAR RD.

LINCOLN RD.

JEFFERSON RD.

HATTON RD.

BROWNS RD.

ASHARD RD.

ADAMS RD.

HARDING RD.

COOLIDGE RD.

JACKSON AVE.

FINLEY LAKE AVE.

OLD STATE AVE.

BRINGOLD RD.

HEMLOCK RD.

HICKORY RD.

JACKSON RD.

ANN ARBOR R.R.

1,148 ft.

1,181 ft.

1,213 ft.

1,246 ft.

1,279 ft.

1,312 ft.

1,115 ft.

1,082 ft.

1,049 ft.

984 ft.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP

Lake Superior

Lake Huron

Lake Michigan

Lake Erie

MICHIGAN

LINCOLN TOWNSHIP

LAKE GEORGE AREA

1ST ST.

PINE ST.

W. SHINGLE LAKE DR.

EVART DR.

LAKE GEORGE RD.

LINCOLN ST.

ARBOR DRIVE

FOREST AVE.

Map 2
Topography

Legend

Elevation Model

- High Elevations
- Middle Elevations
- Low Elevations
- 33 Foot Contour Lines
- Slopes Greater Than 10%

Topography Source: United States Geological Survey (USGS)

Basemap Source: Clare County Equalization Department

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MASTER PLAN
LINCOLN TOWNSHIP
CLARE COUNTY, MICHIGAN

WADE-TRIM
3933 Monitor Road
Bay City, Michigan 48706
(989) 686-3100

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9/17/02

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

LILLY LAKE

MONROE RD.

HARDING RD.

ARTHUR RD.

SHINGLE LAKE

MANNSIDING RD.

JACKSON AVE.

LAKE GEORGE

CEDAR RD.

COOLIDGE RD.

FINLEY LAKE AVE.

OLD STATE AVE.

BRINGOLD RD.

ASHARD RD.

BUNGO LAKE

LINCOLN RD.

BERTHA LAKE

HEMLOCK CR.

JEFFERSON RD.

HEMLOCK LAKE

KILBORNE LAKE

ANN ARBOR R.R.

HATTON RD.

PERCH LAKE

SILVER LAKE

HEMLOCK RD.

HICKORY RD.

BROWNS RD.

BRINGOLD RD.

JACKSON RD.

HALFMOON LAKE

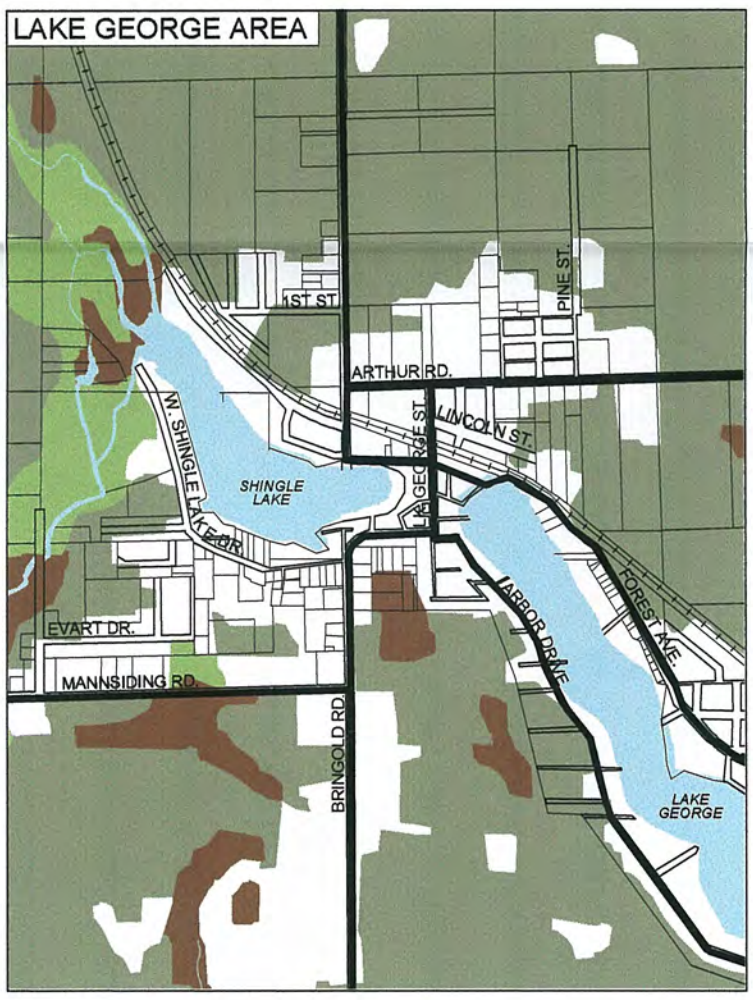
ADAMS RD.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP



LAKE GEORGE AREA



Map 3 Environmental Resources

Legend

- Upland Forest
- Lowland Forest
- Wetland
- Water

Environmental Source: Michigan Resource Information System (MIRIS), Michigan Department of Natural Resources (MDNR)

Basemap Source: Clare County Equalization Department

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9/17/02

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

LILLY LAKE

MONROE RD.

HARDING RD.

ARTHUR RD.

SHINGLE LAKE

MANNSIDING RD.

JACKSON AVE.

LAKE GEORGE

CEDAR RD.

COOLIDGE RD.

FINLEY LAKE AVE.

Primark Property

BRINGOLD RD.

ASHARD RD.

Great Lakes Gas Transmission Co.

BUNGO LAKE

OLD STATE AVE.

LINCOLN RD.

HEMLOCK CR.

BERTHA LAKE

JEFFERSON RD.

3991 S. Hickory Road

HEMLOCK LAKE

KILBORNE LAKE

ANN ARBOR R.R.

HATTON RD.

108 W. North St.

Oil & Gas Contamination Site

PERCH LAKE

HEMLOCK RD.

SILVER LAKE

HICKORY RD.

BROWNS RD.

JACKSON RD.

BRINGOLD RD.

HALFMOON LAKE

ADAMS RD.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP

Lake Superior

Lake Huron

LINCOLN TOWNSHIP

Lake Michigan

MICHIGAN

Lake Erie

LAKE GEORGE AREA

1ST ST.

PINE ST.

490 Arthur Road

ARTHUR RD.

W. SHINGLE LAKE DR.

LAKE GEORGE ST.

LINCOLN ST.

162 Lake George St.

SHINGLE LAKE

EVART DR.

MANNSIDING RD.

BRINGOLD RD.

ARBOR DRIVE

FOREST AVE.

LAKE GEORGE

Map 4 Soils

Legend

Soil Association

- Montcalm-Menominee-Nester
- Menominee-Iosco-Kawkawlin
- Graycalm-Montcalm
- Grayling
- Rubicon-Croswell-AuGres
- Lupton-Markey

Leaking Underground Storage Tank (LUST) Site

Site of Environmental Contamination

Water

Soil Association Source: U.S. Department of Agriculture Soil Conservation Service.

Contamination Sites Source: Michigan Department of Natural Resources.

LUST Sites Source: Michigan Department of Environmental Quality.

Basemap Source: Clare County Equalization Department

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LINCOLN TOWNSHIP
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9/17/02

5.0 Existing Land Use Analysis

The rational application of the planning process for the Future Land Use Plan is possible only when there is a clear understanding of existing conditions and relationships between land uses. Knowledge of existing land development furnishes the basic information by which decisions can be made concerning proposals for future residential, commercial, industrial, and public land use activities. The Existing Land Use Map and Table, included in this section of the report, will serve as a ready reference for the Township in its consideration for land use management and public improvement proposals.

5.1 Survey Methodology

A computer-generated base map for the Township was first created using digital information from the Michigan Resource Information System (MIRIS), and the Clare County Equalization Department. The base map includes the Township boundary line, streets, water bodies and channels, and parcel lines. A parcel-by-parcel field survey of the entire Township was conducted by Wade-Trim in April 2002. Each land use was recorded on the base map according to a predetermined land use classification system. The information was later digitized and, thus, the Existing Land Use Map was created (**Map 5**). The map was reviewed with the Township officials for accuracy. Land use acreages were then derived directly from the digital information (**Table 13**).

Table 13
Existing Land Use Acreage
Lincoln Township, 2002

Land Use Category	Acres	% of Total
1. Single-Family Residential (includes Small-Track Residential)	4,665	20.3
2. Commercial/Office	33	0.1
3. Industrial	85	0.4
4. Property Owned by Gas Companies	3,232	14.1
5. Public/Semi-Public	23	0.1
6. Recreation	108	0.5
7. State Land	183	0.8
8. Vacant/Right-of-Way/Other	14,057	61.2
9. Water	581	2.5
Totals	**22,967**	**100.0**

Source: Wade-Trim Field Survey of April, 2002 with subsequent corrections.

5.2 Land Use Analysis

The existing land use analysis describes the current land uses found within the Township. The overall land use pattern in the Township can be characterized as rural. Although pockets of residential land uses are found in the Township, the majority of the Township consists of vacant wooded lands. The largest land use in the Township is Vacant/Right-of-Way/Other, accounting for over 60% of the Township. Each land use category in the Township is described below.

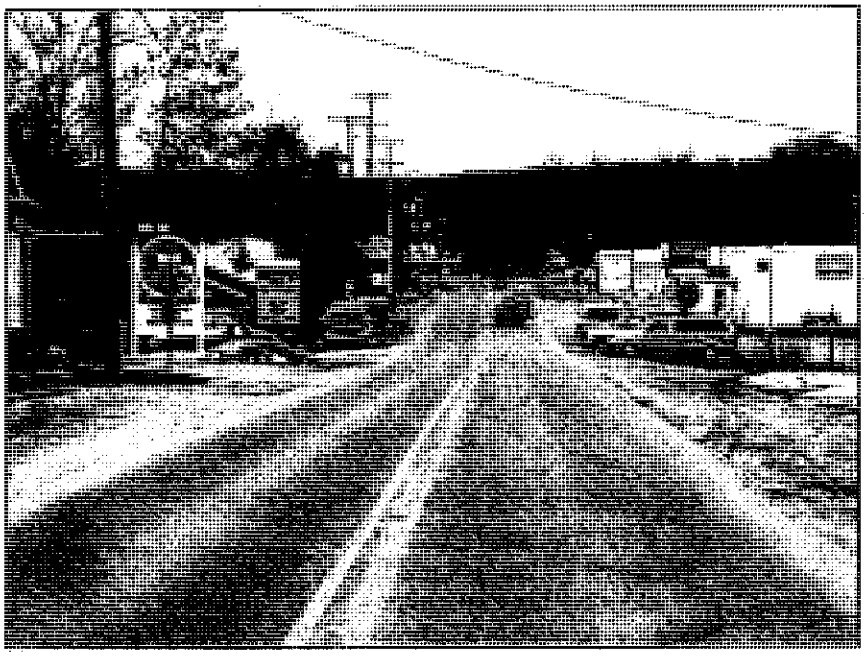
1. Single-Family Residential

   This land use category includes land occupied by single and two-family dwelling units, seasonal dwellings, mobile homes outside of designated mobile home parks, and their related accessory buildings such as garages and sheds. For the purpose of this land use description, the small track residential category as shown on **Map 5** is included in this category. Small track residential areas are also comprised of single-family, two-family, or individual mobile homes, but have lots too small to show clearly on the map.

   The majority of the single-family housing in Lincoln Township is densely concentrated around several lakes including Bertha Lake, Lake George, Shingle Lake, and Silver Lake. In general, these lakefront areas are composed of older cottage style homes on small lots, many of which are occupied by seasonal residents and vacationers. Large modern homes are less common but can be found scattered within these lakefront residential areas.

   Another large residential development is the private White Birch Lakes Subdivision located off of Old State Avenue. This gated subdivision occupies a land area of well over two square miles and features homes interspersed within heavy woodlands.

   The rest of the single-family homes are scattered throughout the Township. In total, single-family residential uses comprise 4,665 acres or 20.3% of the Township. Currently, no multiple-family uses are found in Lincoln Township.



Lake George Commercial District

2. Commercial/Office

   This land use category includes land that is predominantly occupied by retail sales or service establishments. Offices such as financial institutions, professional offices, and clinics are also included in this category.

   Commercial land uses account for only 33 acres of the Township. Most of the commercial establishments in Lincoln Township are concentrated in the Lake George area. Typical commercial uses in this area include restaurants, party stores, realtors, and auto service stations. Collectively, the commercial uses in this business district provide for

many of the everyday commercial needs of the citizens of Lincoln Township.

Other commercial districts within a short distance of Lincoln Township are located in the communities of Clare, Farwell, and Harrison. Regional shopping centers, although not available in the immediate area, are located within easy driving distance in the larger communities of Cadillac, Midland, Mount Pleasant, and Saginaw.

3. Industrial

This land use category is comprised of land occupied by manufacturing industries, processing facilities, warehouses, and non-manufacturing uses, which are primarily industrial in nature. Lands so classified may include areas with or without buildings where raw or semi-finished materials are fabricated or those using or storing raw materials for primary production or extractive operations such as mining sites.



Great Lakes Gas Transmission Company

Several industrial establishments are currently found in the Township, accounting for a total of 85 acres or 0.4%. The two primary industries are the ANR Pipeline Company and Great Lakes Gas Transmission Company, both gas processing facilities located on Hickory Road.

4. Property Owned by Gas Companies

Lincoln Township currently has a large amount of property being used for gas extraction purposes. Because of this, a special category for property owned by gas companies has been added to the Existing Land Use Map. This category includes all lands under gas company ownership whether they are currently being used for gas extraction or are vacant. In total, this category comprises a significant percentage of the Township at 3,232 acres or 14.1%.

5. Public / Semi-Public

Public uses are land and facilities that are publicly operated and available for use by the public. Examples include schools, government buildings, water and sewer utilities, correctional facilities, hospitals, and airports. Semi-public uses are land and facilities which may be privately owned or operated but used by the public or a limited number of persons. Examples include churches, cemeteries, and power line rights-of-way.

Public and semi-public uses comprise 23 acres or 0.1% of the Township. All located in the Lake George Area, these public and semi-public uses include the Lake George Community Church, Lincoln Township Fire Department, Lincoln Township Cemetery, Lincoln Township Hall, Lincoln Township Compost Site, and a United States Post Office.

6. Recreation

Recreation facilities including parks, marinas, private outdoor activity clubs, and golf courses are included in this category. In total, the Recreation category comprises 108 acres or 0.5% of the Township. A description of the recreation facilities in the Township is included below:

- **Bertha Lake Park** is located on Bertha Lake. The 1-acre park is owned by the Bertha Lake Property Owners Association and maintained by Lincoln Township. The park features a pavilion, restrooms, and a non-motorized boat launch.
- **Lake George Campground** is a private campground located near Lake George on Cedar Road. This approximately 20-acre facility contains 65 campsites with a pavilion, volleyball court, and basketball court.
- **Lake George Community Center** is located in the heart of Lake George. The building is also the location of the Township Library and Township Hall. Senior citizens events and other social events frequently are held at this facility.
- **Luke Hamlin Field** is located on Bringold Road south of the community of Lake George. The 25-acre park contains one pavilion with no picnic tables, two baseball fields, one basketball court, two slides, and some jungle gym equipment.
- **Shingle Lake Park** is located on the northern shore of Shingle Lake. The 3-acre park is owned by the Township and features two pavilions, restrooms, playgrounds, picnic tables, beach area, and a boat launch.
- **Silver Lake Park** is located on the east side of Silver Lake. The Township owned park features one pavilion, restrooms, playgrounds, and a boat launch on two acres.
- **White Birch Lake Campground** (private) is located within the White Birch Lakes Subdivision on Old State Avenue in the southeastern portion of the Township. The campground occupies 57 acres.

7. State Land

Lands that are owned and operated by the State of Michigan for purposes such as conservation or research are included in this category.

The State of Michigan owns 183 acres or 0.8% of the Township. This state land is found in two locations: Northwest of Shingle Lake and surrounding Lilly Lake.

8. Vacant/Right-of-Way/Other

Vacant lands include undeveloped forest land, wetlands, and barren lands. All highway, road, and railroad rights-of-way are also included in this category. Agricultural uses such as crop production and grazing are also included in this category.

In total, this category comprises the largest amount of Lincoln Township at 14,057 acres or 61.2%. The vast majority of the lands in this category are vacant and heavily wooded. There are only a few parcels currently being utilized for agricultural purposes in the Township.

9. Water

Lincoln Township is a community of lakes. The lakes are a primary contributor to the Township's overall desirability and attractiveness. In total, lakes comprise 581 acres or 2.5% of the Township. Some of the larger lakes in the Township are:

Name	Approximate Size
Bertha Lake	43 Acres
Halfmoon Lake	55 Acres
Hemlock Lake	22 Acres
Lake George	134 Acres
Kilborne Lake	73 Acres
Perch Lake	28 Acres
Shingle Lake	33 Acres
Silver Lake	50 Acres

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

LILLY LAKE

MONROE RD.

HARDING RD

ARTHUR RD.

SHINGLE LAKE

JACKSON AVE.

MANNSIDING RD.

LAKE GEORGE CAMPGROUND

COOLIDGE RD.

LAKE GEORGE

CEDAR RD.

FINLEY LAKE AVE.

BRINGOLD RD.

ASHARD RD.

BUNGO LAKE

OLD STATE AVE

LINCOLN RD.

GREAT LAKES GAS TRANSMISSION CO. STATION 12 LAKE GEORGE

ANR PIPELINE COMPANY LINCOLN STATION

BERTHA LAKE PARK

BERTHA LAKE

HEMLOCK CR.

JEFFERSON RD.

WHITE BIRCH CAMPGROUNDS (PRIVATE)

HEMLOCK LAKE

KILBORNE LAKE

SILVER LAKE PARK

ANN ARBOR R.R.

HATTON RD.

WHITE BIRCH LAKES SUBDIVISION (GATED PRIVATE ASSOC.)

PERCH LAKE

SILVER LAKE

HEMLOCK RD.

HICKORY RD.

BROWNS RD.

BRINGOLD RD.

JACKSON RD.

HALFMOON LAKE

ADAMS RD.

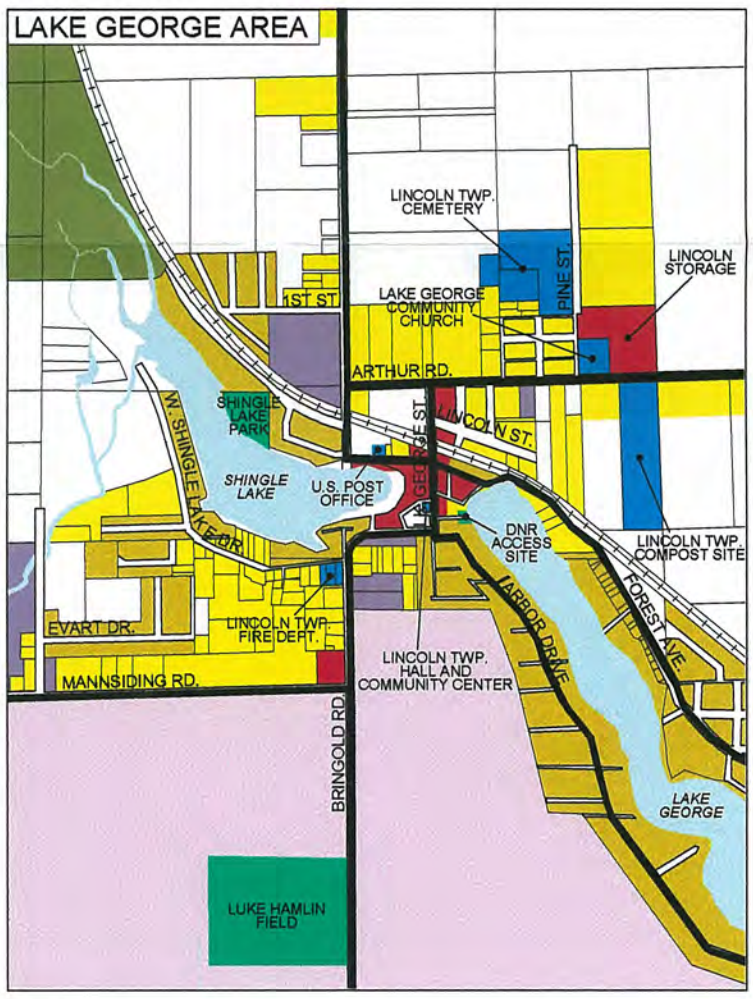
WHITE BIRCH LAKES SUBDIVISION (GATED PRIVATE ASSOC.)

1/2 1/4 0 1/2 1 Mile

LOCATION MAP



LAKE GEORGE AREA



Map 5
Existing Land Use

Legend

- Single-Family Residential
- Small Track Residential
- Commercial/Office
- Industrial
- Property Owned By Gas Companies
- Public/Semi-Public
- Recreation
- State Land
- Vacant/Right-of-Way/Other
- Water

Land Use Source: Wade-Trim Field Survey of April 2002 with subsequent corrections

Basemap Source: Clare County Equalization Department

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MASTER PLAN
LINCOLN TOWNSHIP
CLARE COUNTY, MICHIGAN

WADE-TRIM
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6.0 Transportation Analysis

Mobility constitutes a vital part of the social and economic well being of a community. The traffic circulation system is, in a large sense, the framework upon which the Township is built. This system must support the collective mobility of citizens and visitors of Lincoln Township. The system must also be fully coordinated with the other elements of the master plan, particularly future land use, so as to complement the collective goals, objectives, and policies of the Plan.

6.1 Transportation Network

The residents of Lincoln Township have good access to the region and state through major county roads, local roads, nearby State and U.S. Highways, and other forms of transportation.



Bringold Road

Although no State or U.S. Highway passes through the limits of Lincoln Township, several are located just outside. To the north is M-61, to the west is M-115, to the south is U.S. 10, and to the east is U.S. 127.

These highways provide easy access to the larger communities in the region such as Cadillac, Clare, and the Tri-cities of Bay City, Midland, and Saginaw. These highways also provide access to the major metropolitan areas of the state as well as the entire Midwest. Major cities located within a one-half day driving distance from the Township include:

City	Miles
Chicago	275
Cleveland	310
Detroit	160
Grand Rapids	100
Indianapolis	340
Lansing	90
Toronto	340

Map 6 shows the current transportation network of Lincoln Township. All roads in the Township are divided into three categories: County Primary Road, County Local Road, and Other Local Road. The map also shows the location of railroads in the Township.

Several county primary roads are located in the Township including Mannsiding Road, Bringold Road, Cedar Road, and Old State Avenue. These County Primary Roads are managed by the Clare County Road Commission and provide access within Lincoln Township and to the surrounding communities. Although the Township has a significant amount of county primary roadways, some of the county primary roads are unpaved, while many of the other primary roads follow indirect routes with numerous jogs and 90 degree turns. These characteristics create a somewhat

fragmented system of county primary roads making the Township less accessible and confusing to visitors.

County Local Roads are also managed by the Clare County Road Commission but mainly serve as local access roads within the Township. The rest of the roads in Lincoln Township are classified as Other Local Roads. These roads are less traveled and generally serve only the residents that live along them.

Public transportation is available to Lincoln Township residents through the Clare County Transit bus line. Airline travel is provided through three airports. The Clare County Airport is located in the Harrison Area while the Clare Municipal Airport is located outside of Clare. The nearest commercial airport is MBS (Midland, Bay City, Saginaw) International Airport in Saginaw.

6.2 Transportation Analysis

Map 7 shows a variety of data relating to the current condition of roads within Lincoln Township. All of the roads have been classified as either paved or unpaved. Information on paved and unpaved roads was gathered through a survey conducted by Wade-Trim in April of 2002.

Most of the roads in Lincoln Township are unpaved. Only a select number of roads are paved including Mannsiding Road leading west from Lake George, Bringold Road heading north from Lake George, and Arthur/Jackson/Cedar/Finley Lake/ Jefferson/Old State Roads connecting Lake George with U.S. 10 and Farwell to the south. Currently, no paved road provides a connection between Lake George and areas to the east of the Township including the U.S. 127 Freeway.

Map 7 also shows the scheduled road improvements for Lincoln Township as determined by the Clare County Road Commission. As can be seen, several major projects are scheduled for improvement which should help improve the overall transportation network of the Township.

Map 7 also identifies dangerous intersections within Lincoln Township as determined by Wade-Trim during the land use survey of April 2002. Five dangerous intersections were found:

- Mannsiding Road at Bringold Road
- Finley Lake Avenue at Jefferson Road
- Lake George Street at Arbor Drive
- Cedar Road at Finley Lake Avenue
- Old State Avenue at White Birch Lakes Subdivision Entrance

Traffic safety issues could arise at these intersections due to high speeds, sharp curves, excessive traffic, or driver confusion.

Average annual daily traffic (AADT) counts on selected roads are also shown on **Map 7**. The data for traffic counts was obtained through the Clare County Road Commission. Knowing traffic volumes is important for identifying the corridors that generate the highest traffic and visibility, and which have a greater potential for supporting more intensive land uses. Traffic volume count is also used for determining the need for road improvement and expansion.

Overall, the traffic volumes on Lincoln Township's roads are low. The highest AADT count of 1,887 is found on Lake George Street in the community of Lake George. Several other AADT counts of 900 or more are also found in the Lake George area.

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

LILLY LAKE

MONROE RD.

HARDING RD.

ARTHUR RD.

SHINGLE LAKE

MANNSIDING RD.

JACKSON AVE.

COOLIDGE RD.

LAKE GEORGE

CEDAR RD.

FINLEY LAKE AVE.

BRINGOLD RD.

LASHARD RD.

BUNGO LAKE

OLD STATE AVE.

LINCOLN RD.

BERTHA LAKE

HEMLOCK CR.

JEFFERSON RD.

HEMLOCK LAKE

KILBORNE LAKE

ANN ARBOR R.R.

HATTON RD.

PERCH LAKE

SILVER LAKE

HEMLOCK RD.

HICKORY RD.

BROWNS RD.

BRINGOLD RD.

JACKSON RD.

HALFMOON LAKE

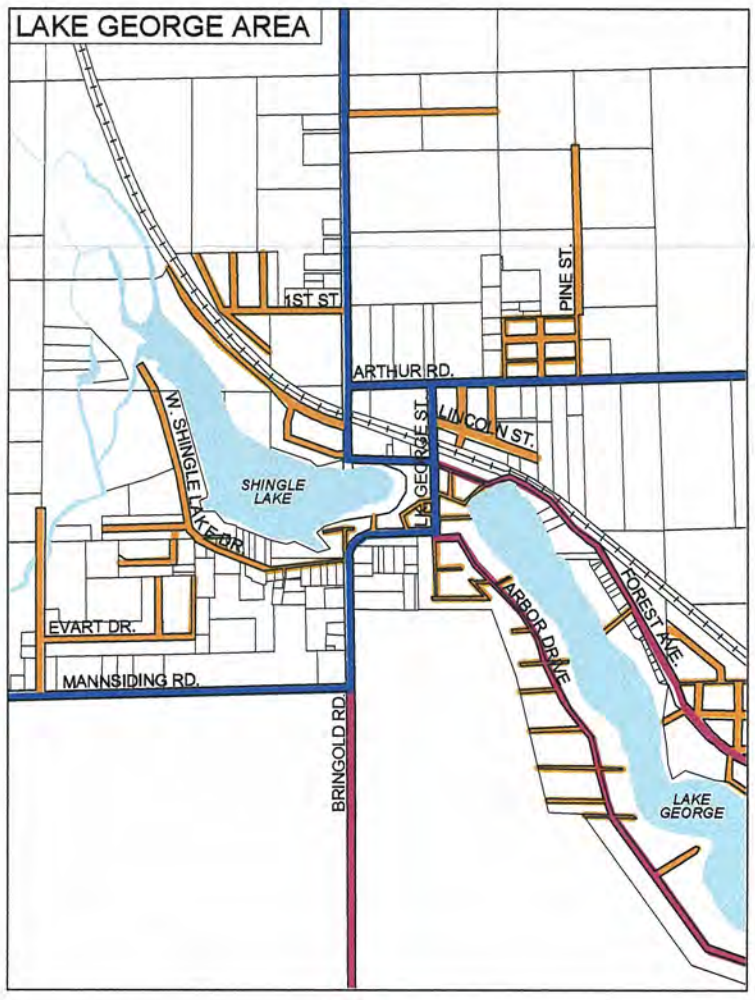
ADAMS RD.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP



LAKE GEORGE AREA



Map 6 Transportation Network

Legend

- County Primary Road
- County Local Road
- Other Local Road
- Railroad

Transportation Source: Clare County Road Commission

Basemap Source: Clare County Equalization Department

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LINCOLN TOWNSHIP
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GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

LILLY LAKE

746 '97

2003/4 REPAVE

2008/9 PAVE

SEE INSET

MONROE RD.

HARDING RD

ARTHUR RD.

SHINGLE LAKE

JACKSON AVE.

COOLIDGE RD.

2004/5 PAVE

MANNSIDING RD.

554 '87

339 '87

LAKE GEORGE

CEDAR RD.

1,010 '82

FINLEY LAKE AVE.

2006/7 PAVE

BRINGOLD RD.

ASHARD RD.

BUNGO LAKE

OLD STATE AVE.

LINCOLN RD.

2002/3 PAVE

2003/4 PAVE

HEMLOCK CR.

BERTHA LAKE

795 '82

JEFFERSON RD.

HEMLOCK LAKE

KILBORNE LAKE

ANN ARBOR R.R.

HATTON RD.

PERCH LAKE

HEMLOCK RD.

SILVER LAKE

HICKORY RD.

989 '82

BROWNS RD.

BRINGOLD RD.

JACKSON RD.

2003/4 REPAVE

HALFMOON LAKE

ADAMS RD.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP

Lake Superior

Lake Huron

LINCOLN TOWNSHIP

Lake Michigan

MICHIGAN

Lake Erie

LAKE GEORGE AREA

922 '93

1ST ST

PINE ST.

ARTHUR RD.

W. SHINGLE LAKE RD.

SHINGLE LAKE

LINCOLN ST.

1,375 '87

1,887 '97

1,554 '97

EVART DR.

FOREST AVE.

MANNSIDING RD.

947 '93

539 '97

ARBOR DRIVE

LAKE GEORGE

BRINGOLD RD.

396 '82

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Map 7 Transportation Analysis

Legend

- Paved Road
- Unpaved Road
- Dangerous Intersection
- Scheduled Improvement with Year
- 795 '82 AADT* with Year

* Average Annual Daily Traffic (AADT) count taken during indicated year.

Transportation Source: Clare County Road Commission & Wade-Trim field survey of April 2002

Road Improvements: Clare County Road Commission

Basemap Source: Clare County Equalization Department

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9/17/02

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7.0 Community Goals and Objectives

7.1 Introduction

Before a community can actively plan for its future growth and development, it must first establish a vision for its future in the form of a set of goals and objectives. Goals and objectives define the boundaries of the community's needs and aspirations. They must reflect the type of community desired by its citizens, given realistic economic and social limitations.

The following is a recommended set of community goals (the ultimate desire of the community, or the intent of the Master Plan) and objectives (the means of attaining goals). These goals and objectives are based upon the Background Information Analysis, as presented in Sections 3.0 through 6.0, and upon a survey of the citizens and Planning Commission Members conducted on May 21, 2002.

7.2 Community-Wide Goals

1. Preserve and protect the Township's abundant natural resources such as woodlands, lakes, and wetlands.

2. Capitalize on the Township's abundant resources (i.e. natural resources, central location, and close proximity to US-27, US-10, M-115, and M-61) for the purpose of creating an attractive community that meets the physical, social, and economic needs of its residents and businesses in an environmentally sensitive manner.

3. Manage new growth in a manner that will retain the rural character of the community.

4. Preserve and promote the rights of individual property owners while maintaining the aesthetic character of the community.

5. Relate land use primarily to the natural characteristics of the land and the long-term needs of the community, rather than to short-term economic gain.

6. Encourage intergovernmental cooperation between Lincoln Township, surrounding communities, Clare County, and the State of Michigan in the provision of area-wide facilities.

7. Alleviate blight to present a better image of the Township.

8. Provide for the special needs of the rapidly growing elderly population in the Township.

7.3 Residential Goal and Objectives

Goal

Provide a wide variety of safe, attractive, sanitary, and affordable housing choices to present and future residents of Lincoln Township.

Objectives

1. Encourage the improvement and beautification of existing residences as a means of ensuring lasting identity and stability of residential areas.

2. Encourage new residential developments whose densities, styles, and locations will not adversely effect the current rural and small town character of the community.

3. Discourage residential development in environmentally sensitive areas.

4. Require adequate buffers or transition areas between residential and

non-residential developments to maintain property values and attractiveness.

5. Encourage the alleviation of conflicting or undesirable land uses from residential areas through code enforcement and other options.

6. Encourage the alleviation of unsanitary or unsafe housing through code enforcement or other options.

7. Encourage new multi-unit and group housing developments to meet the needs of elderly population.

7.4 Commercial Goal and Objectives

Goal

Provide for a selected range of commercial facilities to serve the needs of area population.

Objectives

1. Encourage the development of small commercial establishments in strategic locations.

2. Discourage strip commercial development along highways, except where a specific need can be substantiated for highway-oriented type businesses and where such businesses will not adversely impact existing residential uses.

3. Establish a compatible relationship between commercial and adjacent residential uses through the use of buffer devices such as walls, fences, landscaped areas, and transitional uses.

7.5 Industrial Goal and Objectives

Goal

Encourage a variety of light industrial developments with attractive sites to strengthen the tax base and to provide employment opportunities for area residents.

Objectives

1. Encourage the selective development of new industries that are economically associated with the existing industrial base in the region.

2. Locate industrial areas that have reasonable boundaries, are easily accessible from the existing transportation network, and are not subject to encroachment by incompatible uses.

7.6 Transportation Goal and Objectives

Goal

Develop and maintain a network of paved, all weather roads that meets the needs of all Township residents and businesses in a safe and convenient manner.

Objectives

1. Utilize the federal/state/county road and highway classification system for classifying existing and future roads in Lincoln Township.

2. Cooperate with the Clare County Road Commission in the reduction of fragmentation of the county primary road system within the Township.

3. Limit points of ingress/egress on major roads.

4. Segregate truck, automobile, and off-road trail traffic as much as possible.

5. Develop and implement a plan for the improvements of unpaved roads through a public participation process.

6. Improve traffic safety by improving five dangerous intersections identified on **Map 7** of this report.

7. Develop and implement a transportation beautification plan focusing on streetscape, signage, landscaping, and other aesthetic improvements along major roads within the Township.

8. Comply with the provisions of the Americans with Disabilities Act (ADA) in all new transportation facilities.

7.7 Infrastructure Goal and Objectives

Goal

Meet the infrastructure needs of residents and businesses in a cost-effective and environmentally sensitive manner.

Objectives

1. Conduct a feasibility study for the provision of public sanitary sewer and water systems in the Township.

2. Cooperate with surrounding communities in meeting infrastructure needs of the community.

3. Match new development with the available or planned infrastructure capacities.

7.8 Park and Recreation Goal and Objectives

Goal

Initiate an aggressive program to develop a wide variety of recreational facilities to meet the needs of all segments of the Township population.

Objectives

1. Develop a State-certified Community Recreation Plan to identify local recreation needs and seek State funding to fulfill such needs.

2. Develop a Township-wide network of bike trails to connect major recreation facilities in the Township with the residential neighborhoods.

3. Promote the development of new recreation facilities that will meet the recreation needs of the elderly population.

7.9 Natural Environment Goal and Objectives

Goal

Preserve and enhance the natural and environmental resources of the Township for all current and future Township residents.

Objectives

1. Seek planning and engineering solutions to improve the shoreline development along the major lakes for the purpose of protecting the water quality, controlling damage from flooding, and providing public access to the lakes.

2. Implement land use patterns, which will direct new growth away from environmentally sensitive areas such as

woodlands, wetlands, steep slopes, and areas subject to flooding.

3. Implement development controls, which will maximize the protection of land-based natural resources while preserving the quality of air and water.

4. Encourage and seek solutions to protect wetlands and groundwater resources.

8.0 Future Land Use Plan

8.1 Introduction

The Future Land Use Plan identifies the desired pattern of land development in Lincoln Township for a period extending approximately 20 years. The Future Land Use Plan is a general statement of the Township's goals and provides a single, comprehensive view of the community's desire for its future. This section describes the basis for the plan and the intended character of each land use classification.

8.2 What is a Future Land Use Plan?

A Future Land Use Plan is a guide to the physical development of a community. Based on the social and economic values of the community, it translates those values into a scheme that describes how, why, when, and where to build, rebuild, or preserve the community.

There are many general characteristics of a Future Land Use Plan. The first characteristic of the Plan is that it is long range, covering a time period of as much as twenty years. Not only does the Plan present a vision of the community in the future, it also recommends procedures and policies that can help the community to get there.

A second characteristic of the Plan is that it is meant to be general in nature. The Future Land Use Plan is not meant to be specific, recommending land uses property by property. It only provides land use recommendations for generalized locations in the community. This is one of the strengths of the Future Land Use Plan, allowing for the community to determine exact locations and boundaries for the proper land use classifications.

Lastly, a Future Land Use Plan is a statement of policy and a guide covering such community desires as quantity, character, location, and rate of growth and indicating how these desires are to be achieved. It is important to understand that the Plan has no legal authority, unlike legal documents such as a zoning ordinance or subdivision regulations. The Plan serves as a guide in the formulation of these legal documents. Government bodies and officials such as the Township Board, Planning Commission, and Zoning Administrator should use the Plan as a guide in their day-to-day decision making processes.

8.3 Basis for the Plan

Making informed decisions about the future growth and redevelopment of communities is no easy task. Communities have become centers of complex and interrelated activities. Employment and residential areas are interconnected and supported by public and private facilities such as streets, water, sewer, storm drains, parks, and services such as, garbage pick up, police and fire protection, medical and emergency services, recreation and entertainment, and personal services. Many of these facilities and services are interrelated, as are the land uses they support or serve. A Future Land Use Plan can only be created after this thorough understanding of all the elements in the land use system is gained.

In the proceeding sections of this Master Plan, many elements of Lincoln Township were analyzed including demographic data, environmental features, existing development patterns, existing transportation network, and community goals and objectives. These elements, in addition to an understanding of regional conditions and market trends, combine to form the basis for the Future Land Use Plan.

8.4 Plan Recommendations

Twelve land use classifications are proposed for Lincoln Township. The various future land uses are portrayed on **Map 8** and the total acreages for each category are shown in **Table 14**.

Table 14
Future Land Use Acreage
Lincoln Township

Land Use Category	Acres	Percent
1. Rural Residential	11,849	51.6
2. Single-Family Residential	5,720	24.9
3. Multi-Family Residential	64	0.3
4. Commercial	27	0.1
5. Central Business District	23	0.1
6. Industrial	263	1.1
7. Property Owned by Gas Companies	3,232	14.1
8. Public/Semi-Public	23	0.1
9. Recreation	108	0.5
10. State Land	183	0.8
11. Water	581	2.5
12. Right-of-Way	894	3.9
Totals	**22,967**	**100.0**

A description of each Future Land Use category is presented below.

1. Rural Residential

In keeping with the Community-Wide Goal of "preserving and protecting the Township's abundant natural resources such as woodlands, lakes, and wetlands," the largest percentage (51.6%) of the Township has been classified into the rural residential land use category. The rural residential category is designed to accommodate rural land uses consistent with the Township's existing rural character and natural resources. Lands designated into this category are primarily located in the southern and eastern portions of the Township.

The rural residential category is intended to accomplish a number of goals:

- Preserve the environmental resources, including upland and lowland forests within the Township, for future generations.
- Steer development away from environmentally sensitive areas in the Township such as wetlands.
- Prevent large-scale residential developments from locating in the district.

Low-density single-family homes compatible with the rural character of the Township are welcome in this district. Minimum lot sizes should be one or more acres per housing unit. Other compatible land uses within this rural residential category may include agricultural activities, outdoor commercial recreation activities such as riding stables and golf courses, and

open space uses such as campgrounds and nature preserves.

2. Single-Family Residential

The single-family residential classification is intended to accomplish two major goals: enhance the existing single-family residences in the Township, and encourage new single-family development. Lands for new single-family development are needed in the Township because of the following population trends:

- Over the last 30 years, the population of Lincoln Township grew by more than 170%. New single-family residences will be needed to accommodate this significant population growth.
- The number of persons per household in the Township has declined over the past two decades. Because of this, more homes will be needed to accommodate the fewer number of people living in each home.

The single-family residential classification can be found concentrated in several areas of the Township. The largest concentration is found in the northwest corner of the Township in the Lake George area. Other concentrations are found surrounding Silver Lake, Bertha Lake, and in the existing White Birch Lakes Subdivision. These single-family areas contain both existing residences as well as vacant lands well suited for new residential growth.

Keeping single-family residential development within these designated areas will promote orderly growth, as opposed to "leap-frog development," and will result in a cost-effective provision of Township services.

The single family residential classification, 5,720 acres or 24.9% of the Township, is intended to provide areas for suburban-style, single-family residential development and related uses such as golf courses, athletic fields, churches, and day-care facilities. It is recommended that the district maintain a density of 2 to 4 units per acre for lots with access to public utilities and 1 to 2 units per acre for lots with no public utilities. New single-family developments should conform to the Township's rural nature and be compatible with existing neighborhoods.

3. Multi-Family Residential

The multi-family residential classification is intended to provide opportunities for affordable housing and alternatives to traditional single-family homes. Included in this district are duplexes, townhouses, apartments, mobile home parks, and related uses.

Two new locations for multi-family development are located in the Lake George area, one on Bringold Road, and the other on Arthur Road. The designation of lands for multi-family use is needed in the Township to accommodate population groups such as the elderly (currently comprising 19.9% of the Township), single persons (currently comprising 25.8% of the Township), low income citizens, and citizens preferring alternative housing options other than single-family homes.

In the event that more lands are needed for multi-family uses, the following criteria should be used for evaluating the development and its location:

- In keeping with the small town character of the Township, the density for new developments should be kept to no more than 20 housing units per acre.
- Since multi-family development generates more traffic than single-family development, it should be

located along or in the vicinity of a major thoroughfare.

- The new development should be used as a zone of transition or buffer between single-family uses and commercial or office uses.

4. Commercial

This category is intended to support predominantly freestanding commercial and office uses that serve both the local and regional market. In total the commercial land use category comprises 27 acres of the Township.

All of the existing commercial uses in the Township have been included in this category.

In the event that new commercial uses are needed in the Township, the following criteria should be used to determine the suitability for a new commercial development:

- The local or regional market must show a clearly defined need for the proposed commercial development.
- The development should be located on a major road which will have minimal negative traffic impact on the surrounding properties.
- The proposed development should be compatible with the surrounding property uses and an acceptable buffer or screen should be provided between the proposed development and adjoining residential uses.

5. Central Business District

In order to promote compact commercial growth to serve the needs of the citizens and visitors of Lincoln Township, the plan proposes the creation of a well-defined central business district to be located in and around the current businesses in the heart of the community of Lake George. In total, the central business district comprises 23 acres.

The district will be developed with a theme unique to Lake George and promoted as a regional destination for shopping, service, and recreation. A detailed development plan for the district should be prepared which would encourage the strengthening of existing businesses and addition of new uses. The district would encourage compatible mixed uses, bike, and pedestrian traffic, and unique design controls for building facades and signs.

The unique theme and attractiveness of the district can be established through landscaping features such as trees and flower beds, information booths, and street furniture, such as benches, street signs, light poles, banners, and waste receptacles.

6. Industrial

The Future Land Use Plan designates a total of 263 acres of the Township as Industrial. The category includes existing industries as well as vacant lands well suited for new industrial developments.

These new industrial locations can be found east of the Lake George area along Arthur Road and Jackson Avenue.

Providing new locations for industries is important to Lincoln Township in order to promulgate the established industrial goal of "encouraging a variety of light industrial development with attractive sites to strengthen the tax base and provide employment opportunities for area residents." The following criteria should be used for evaluating new industries:

- The development should have good access via a major road.

- The development should be adequately served by public utilities.
- The proposed development should have no appreciable impact on existing residential neighborhoods.

7. Property Owned by Gas Companies

   This future land use category includes all of the lands currently owned by gas companies, which in total comprises 3,232 acres or 14.1% of the Township. No new lands have been classified into this category.

8. Public/Semi-Public

   The Public/Semi-Public district, 23 acres, is comprised of all the existing public and semi-public uses in the Township. Future public and semi-public uses should be appropriately located in areas suitable for development with adequate utilities and transportation access.

9. Recreation

   This Master Plan has been prepared in conjunction with the preparation of a Community Recreation Plan for Lincoln Township. Prepared through extensive citizen input and participation, the Recreation Plan offers a detailed 5-year Action Plan for recreation improvements in Lincoln Township. The major components of the Action Plan include improvements to Shingle Lake Park, improvements to Luke Hamlin Field, and the development of a bike trail and multi-purpose trail in the Township. The locations of the proposed bike trail and multi-purpose trail are shown on **Map 8**.

10. State Land

    State Land comprises 183 acres or 0.8% of the Township.

11. Water

    Water bodies comprise 581 acres or 2.5% of the Township.

12. Right-of-Way

    Street and railroad rights-of-way comprise 894 acres or 3.9% of the Township.

Additional Recommendations

In addition to the aforementioned land uses, the Future Land Use Plan also recommends the following:

Neighborhood Center

The Plan also designates a general location for a Neighborhood Center at the intersection of Finley Lake Avenue and Jefferson Road. In the future, there may be a need for a local commercial center with establishments that serve the surrounding population. Potential establishments may include local convenience stores or restaurants. The location of these uses should be confined to the intersection and proliferation of larger scale commercial uses is not recommended.

Major and Minor Roads

The Plan classifies all roads as either major roads or local roads. All of the existing County Primary Roads are classified as major roads, with the rest of the roads classified as minor roads. Because of the importance of the major roads as regional thoroughfares, there exists an opportunity for the Township to capitalize through enhancement and beautification of the roads and their adjoining land uses. Streetscape plans, proper landscaping requirements, and access control policies are all strategies that can enhance the experience of travelers on these roads, and create a positive image of the Township.

Township Entryway

In order to present a positive image for people driving into or passing through the Township, the plan recommends the development of four Township entryways.

The locations for the four entryways are as follows:

- Mannsiding Road at the western border of the Township
- Bringold Road at the northern border of the Township
- Mannsiding Road at the eastern border of the Township
- Old State Avenue at the southern border of the Township

An entryway is not intended to be a particular land use district. It is meant to serve as an area where the Township would encourage attractive developments, alleviate blight, if it exists, and install specially attractive features such as ornamental lights, signs, banners, and landscaping. Most of these features can be installed on the street right-of-way, or on private property easements.

Potential Public Utility Service Area

In order to promote compact growth and development, preserve the rural character of the Township, and provide cost-effective public services, the Future Land Use Plan recommends a potential public water and sewer service area. This potential service area is located in the northwest corner of the Township, surrounding the Lake George area where the highest intensity land uses currently exist and are proposed. It is recommended that the Township conducts a feasibility study to further examine the provision of public utilities in this area.

8.5 Buildout Analysis

One of the greatest threats to growing communities is master plans and zoning ordinances that permit, if not encourage, the population to multiply many times over, without the community making a commitment to figure out how to pay the costs associated with servicing this growth. One of the major purposes of this buildout anaysis is to help Lincoln Township determine the optimum population of the Township at the densities proposed in this Plan. Estimating the amount of future population growth will give the Township an insight as to the extent of public services needed to accommodate the new population.

Map 9 divides the land in Lincoln Township into two categories: Non-Buildable Area and Buildable Area. Non-buildable area includes lands with existing uses, lands under the ownership of gas companies, lands under state ownership, rights-of-way, wetlands, and water bodies. These lands have no opportunity to be developed in the future.

Buildable areas are those that are currently vacant and have the potential to be developed in the future. In total, buildable area comprises 13,040 acres of the Township. The buildable area is further divided into two groups, Rural Buildable and Non-Rural Buildable according to the proposed future land use classifications in **Map 8**. For example, buildable area designated as Rural Residential in the Future Land Use Map is shown as Rural Buildable in **Map 9** while buildable area designated as Single-Family Residential in the Future Land Use Map is shown as Non-Rural Buildable in **Map 9**.

Table 15 shows the methodology for determining the optimum population of the Township. The total rural buildable area of Lincoln Township is 10,070 acres. It is estimated that when this land is developed, 10% will be used for non-residential uses such as convenience stores, parks, and churches, and another 10% will be used for the right-of-ways of future streets. This leaves a net rural buildable area of 8,056 acres. If these areas are developed at a density of 1 housing unit per 2 acres of land, the net result would be 4,028 units. Given the current Township persons per household of average of 2.3, the total population would be 9,264 people.

Table 15
Buildout Analysis
Lincoln Township

Total Township Area	**22,967 acres**
Subtract Single-Family Residential Uses	4,665 acres
Subtract Commercial/Office Uses	33 acres
Subtract Industrial Uses	85 acres
Subtract Property Owned by Gas Companies	3,232 acres
Subtract Public/Semi-Public Uses	23 acres
Subtract Recreation Uses	108 acres
Subtract State Land	183 acres
Subtract Rights-of-Way	894 acres
Subtract Water	581 acres
Net Township Area	**13,163 acres**
Subtract wetlands within remaining lands	123 acres
Total Township Buildable Area	**13,040 acres**
Rural Buildable Area	10,070 acres
Non-Rural Buildable Area	2,970 acres
Rural Buildable Area	**10,070 acres**
Subtract 10% for Non-Residential Uses	-1,007 acres
Subtract 10% for Future Right-of-Ways	-1,007 acres
Net Rural Buildable Area	**8,056 acres**
Future Units @ 1 Unit per 2 Acres	4,028 units
Future Population @ 2.3 Persons Per Household	9,264 people
Non-Rural Buildable Area	**2,970 acres**
Subtract 20% for Non-Residential Uses	594 acres
Subtract 10% for Future Right-of-Ways	297 acres
Net Non-Rural Buildable Area	**2,079 acres**
Future Units @ 2 Units per 1 Acre	4,158 units
Future Population @ 2.3 Persons Per Household	9,563 people
Buildout Population Analysis	
Existing Population	1,758 people
New Population (9,264+9,563)	18,827 people
Total Population at Buildout	**20,585 people**

The total non-rural buildable area in Lincoln Township accounts for 2,970 acres. Given 20% devoted for non-residential uses and 10% devoted for future right-of-ways, the net non-rural buildable area would be 2,079 acres. If this land was developed at 2 housing units per acre, the total number of future housing units would be 4,158. Given the Township persons per household average of 2.3, the total population within this non-rural buildable area would be 9,563 people.

If Lincoln Township reaches full buildout, the total population of the Township would be 20,585 people, an increase of 18,827 people from the Township's current population of 1,758. However, it is very unlikely that Lincoln Township will ever reach full buildout because of current market conditions and its location in a predominantly rural area of Central Michigan. The buildout analysis does provide good insight as to extent of public services that may be needed in Lincoln Township in the future. For example, the buildout analysis estimates that in the future, approximately 10,000 people could be living in the non-rural area in the northwest corner of the Township. Knowledge of this gives Lincoln Township the opportunity to prepare for this growth now by making sure that public services will be able to accommodate this population.

8.6 Using the Plan

Now that the Future Land Use recommendations have been laid out for the Township, it is appropriate to describe how to put the Plan to use. It is important to remember that the Master Plan and its Future Land Use Map is not a legal document, and should not be confused with the Zoning Ordinance or map. In fact, the Zoning Map and Future Land Use Map may not even look the same. The key difference is that Zoning deals with land use now, and the Master Plan prescribes a vision for land use in the future and serves as a guide to get there.

One of the principal benefits of having an adopted Master Plan is the foundation it provides for zoning decisions. As the Township Board or Planning Commission is faced with making zoning and land use decisions, the respective bodies should consider the recommendations as set forth in the Master Plan. Rezonings, site plan reviews, and special land uses should conform with the principles found in the Master Plan.

Flexibility is a definite strength of the Master Plan. Changing trends, circumstances, unanticipated opportunities, and unforeseen problems can require an amendment to the Master Plan. If a new development proposal appropriate for the Township does not conform to the Master Plan, the Master Plan should first be amended before the proposal is approved. If an amendment occurs, it is important to know that the rest of the Plan is still relevant. The Plan will only become irrelevant or obsolete if the Master Plan is not updated when changes occur in the municipality.

In order to keep the Master Plan up to date, it is important to schedule periodic reviews of the Master Plan. The recently adopted Public Act 263 of 2001 requires that the Townships look at its existing Master Plan every five years to determine whether any amendments or changes are appropriate for the Township.

GREENWOOD TOWNSHIP

FREEMAN TOWNSHIP

HATTON TOWNSHIP

SURREY TOWNSHIP

SEE INSET

MONROE RD.

HARDING RD.

LILLY LAKE

ARTHUR RD.

SHINGLE LAKE

MANNSIDING RD.

JACKSON AVE

LAKE GEORGE

GOOLIDGE RD.

CEDAR RD.

FINLEY LAKE AVE.

OLD STATE AVE.

BRINGOLD RD.

LASHARD RD.

BUNGO LAKE

LINCOLN RD.

BERTHA LAKE

HEMLOCK CR.

JEFFERSON RD.

HEMLOCK LAKE

KILBORNE LAKE

ANN ARBOR R.R.

HATTON RD.

PERCH LAKE

HEMLOCK RD.

SILVER LAKE

BROWNS RD.

HICKORY RD.

JACKSON RD.

OWL LAKE

BRINGOLD RD.

TOWNLINE LAKE

HALFMOON LAKE

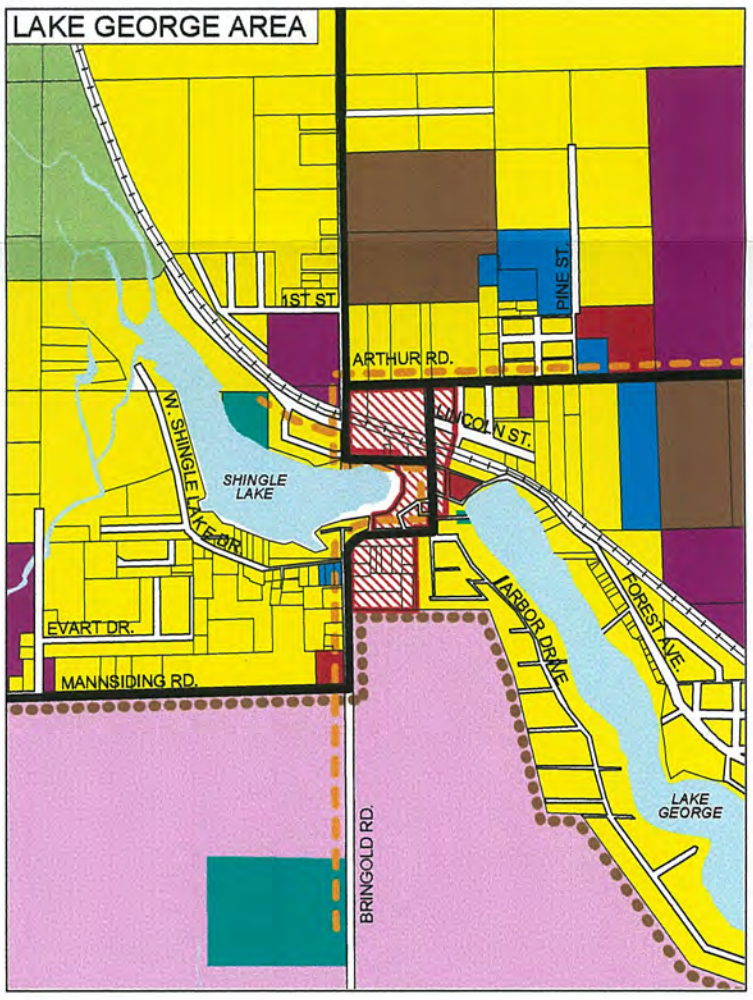
ADAMS RD.

1/2 1/4 0 1/2 1 Mile

LOCATION MAP



LAKE GEORGE AREA



MASTER PLAN
LINCOLN TOWNSHIP
CLARE COUNTY, MICHIGAN

WADE-TRIM
3933 Monitor Road
Bay City, Michigan 48706
(989) 686-3100

"Building Relationships on a Foundation of Excellence"

Map 8
Future Land Use

Legend

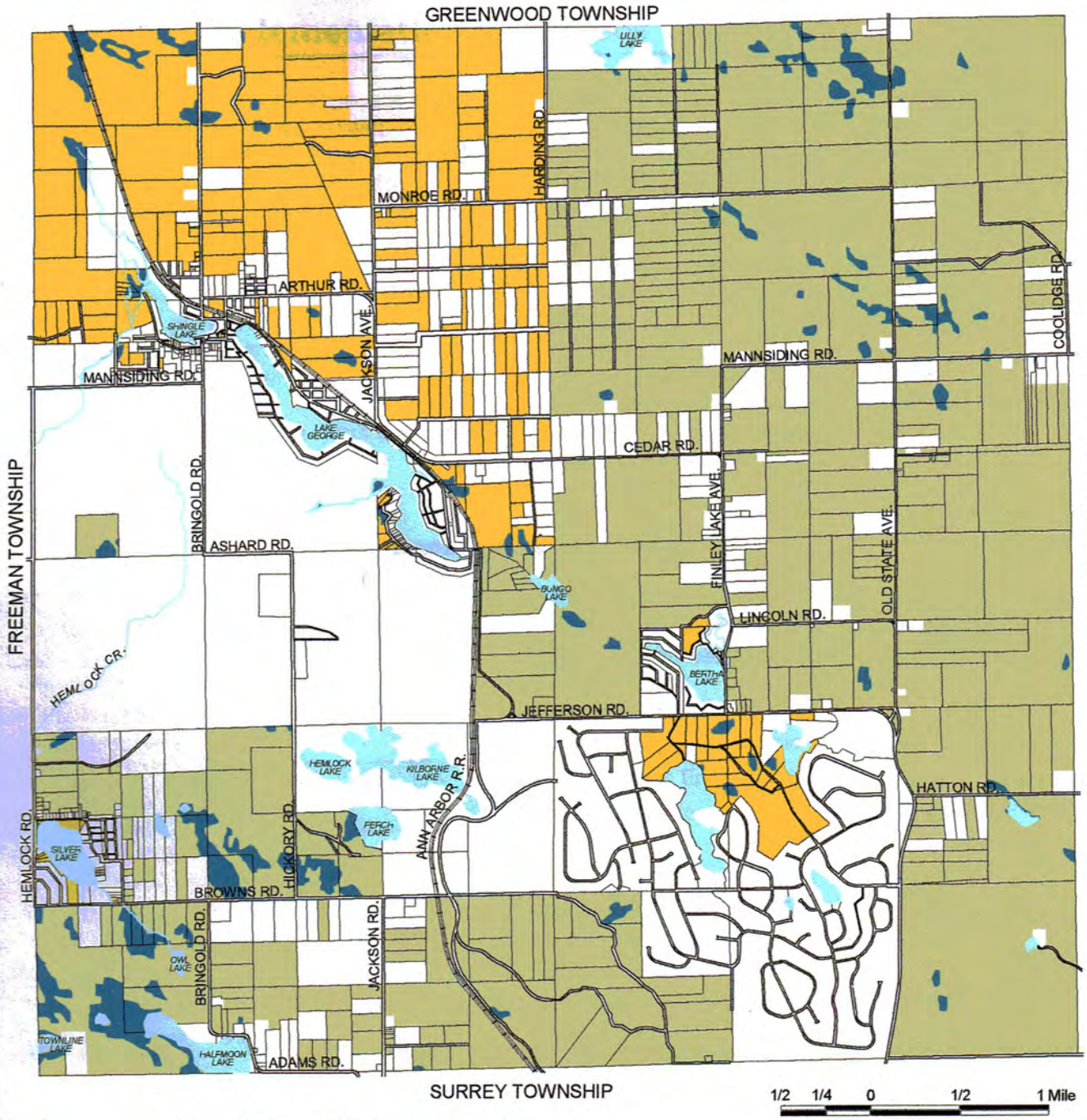
- Rural Residential
- Single-Family Residential
- Multi-Family Residential
- Commercial
- Central Business District
- Industrial
- Property Owned by Gas Companies
- Public/Semi-Public
- Recreation
- State Land
- Water
- Right-of-Way
- Major Road
- Minor Road
- Bike Trail
- Multi-Purpose Trail
- Potential Public Utility Service Area
- Neighborhood Center
- Township Entryway

Basemap Source: Clare County Equalization Department

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9/17/02

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Map 9 Buildout Analysis

Legend

Non-Buildable Area

- Existing Uses, Properties Owned by Gas Companies, Rights-of-Way, and State Land
- Wetlands
- Water

Buildable Area

- Rural Buildable Area
- Non-Rural Buildable Area

Basemap Source: Clare County Equalization Department

MASTER PLAN
LINCOLN TOWNSHIP
CLARE COUNTY, MICHIGAN

WADE-TRIM
3933 Monitor Road
Bay City, Michigan 48706
(989) 686-3100

N

9/17/02

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9.0 Plan Implementation Resources

Lincoln Township's Master Plan is a long-range community policy statement comprised of a variety of both graphic and narrative recommendations intended to provide guidelines for making reasonable and realistic community development decisions. The Plan is intended to be employed by the Township officials, by those making private sector investments, and by all citizens interested in the future development of the Township.

The completion of the Plan is but one part of the community planning process. Realization, or implementation of the goals, objectives, and recommendations contained in the Master Plan can only be achieved over an extended period of time and only through the cooperative efforts of both the public and private sectors. This section will serve as a reference guide for the Township as it moves through the plan implementation process.

Table 16 provides a listing of plan implementation resources included in this section. Following the table is a description of each resource.

Table 16
Plan Implementation Resources

Keys to Successful Implementation	Grant and Loan Programs
1. Knowledgeable Local Officials 2. Public Support 3. Co-Development	**Federal** 1. Community Facilities Program 2. Public Works and Economic Development Program 3. Rural Business Enterprise Grants 4. Water and Wastewater Disposal Loan and Grant Program
Implementation Tools	
1. Zoning Ordinance 2. Subdivision Regulations 3. Enforcement 4. Capital Improvements Program 5. Michigan Certified Community Recreation Plan 6. Downtown Development Authority (DDA) 7. Special Assessment Districts	**State** 5. Community Development Block Grant (CDBG) 6. Land and Water Conservation Fund 7. Michigan Natural Resources Trust Fund 8. Transportation Economic Development Fund 9. Transportation Enhancement Program (TEA-21)
Sources of Information	**Other Funding Options**
1. Michigan Townships Association 2. County Extension Services 3. County Planning Commission 4. Regional Planning Agency 5. Other Local Officials 6. Libraries 7. Internet	**Local** 1. Revenue Bonds 2. General Obligation Bonds 3. Local Funds 4. Local Businesses 5. Local Foundations

9.1 Keys to Successful Implementation

9.1.1 Knowledgeable Local Officials

Ultimately, the responsibility for implementing the Master Plan falls into the hands of the local officials of Lincoln Township. This is why it is very important that the Township Board, Planning Commission, and the various municipal officials should be knowledgeable and focused on achieving the desired results of the Master Plan. The local officials need to be the catalysts for action, leading the community in the right direction.

9.1.2 Public Support

It is critical that the citizens of Lincoln Township acknowledge, understand, and support the Master Plan. In order to organize public support most effectively, Lincoln Township must emphasize the necessity of instituting the Planning Program and encourage citizen participation in the planning process.

Lack of citizen understanding and support could have serious implications for the eventual implementation of planning proposals. Failure of the public to support needed bond issues and continuing dissatisfaction concerning taxation, special assessments, zoning decisions, and development proposals are some of the results of public misunderstanding and rejection of long-range plans.

9.1.3 Co-Development

Implementing the goals and objectives of the Master Plan requires that the private and public sector work together on development projects. Co-development is simply the joint public and private investment for a common purpose. Working in a partnership allows for Lincoln Township to become involved in such things as site location selection, planning, site design, utilities and other service agreements, and tax incentives and abatements. These partnerships help to foster development-friendly environments, where the Township benefits from increased tax revenue, and the private developers can benefit from decreased cost of improvements.

9.2 Implementation Tools

9.2.1 Zoning Ordinance

Zoning Ordinances are essential in implementing the goals and objectives of the Master Plan. The authority to create a Zoning Ordinance is given to a Township by the State under the Township Zoning Act 184 of 1963, as amended, for the purpose of promoting community health, safety, and general welfare. Zoning regulations have been strongly supported by the Michigan courts, as well as by the United States Supreme Court.

The intent of zoning is to assure the orderly development of the community. Zoning does this by dividing the community into districts in order to establish a density of population, and regulate the use of land and buildings. Zoning also promotes the general welfare of a community by protecting homes and investments against the potential harmful intrusion of business and industry into residential neighborhoods, requiring the spacing of buildings far enough apart to assure adequate light and air, preventing the overcrowding of land, facilitating the economical provision of essential public facilities, and aiding in the conservation of essential natural resources. This, in turn, helps to protect the property values of the community.

Because of the importance of zoning, and the great impact which zoning can have on the use of land and related services, it should be based on principles derived from this Master Plan.

9.2.2 Subdivision Regulations

When a developer proposes to subdivide land, he or she is in effect, planning a portion of the Township. To assure that such a development is in harmony with the Master Plan objectives, a subdivision regulation ordinance may be created in accordance with the Michigan Subdivision Control Act 288 of 1967, as amended, and the Michigan Land Division Act 591 of 1996.

Several direct benefits accrue from the regulation of subdivisions by a local unit of government. By requiring the subdivider to install adequate utilities and improved streets, purchasers of the lots are not later burdened with unanticipated expenses. A subdivision without adequate physical improvements is detrimental not only to itself, but it also reduces the opportunity for reasonable development of adjacent parcels. In addition, long-range economy in government can be realized only when the subdivider provides adequate improvements.

As a part of its review of proposed subdivisions, the Planning Commission focuses on such features as the arrangement and width of streets, the grading and surfacing of streets, drainage of storm water, the width and depth of lots, the adequate provision of open space, and the location of easements for utility installations. The subdivision review process is one of the methods of implementing the goals and objectives of the community's long-range Plan.

9.2.3 Enforcement

The ultimate effectiveness of the zoning, subdivision, and other regulations depends on the effective administration and enforcement by the community. If administrative procedures are lax, or if enforcement of regulations is handled in an inconsistent, sporadic manner, the result will be unsatisfactory at best.

The Zoning Administrator is often responsible for carrying out zoning/development related functions, including building inspections, ordinance administration, and community/ developer liaison. Each of these functions requires a substantial investment of staff time. If sufficient time is not made available to carry out these critical functions, they may only be accomplished in a cursory manner. Therefore, the Township should provide for adequate staff levels and/or consulting assistance to assure that these essential day-to-day functions will receive the professional attention required.

9.2.4 Capital Improvements Program

The term "capital improvements" is generally intended to embrace large-scale projects of a fixed nature, the implementation of which results in new or expanded public facilities and services. Such items as public building construction, park development, sewer installation, waterworks improvements, street construction, land acquisition, and the acquisition of certain large-scale pieces of equipment (graders, sweepers, trucks, etc.) are included in the Capital Improvements Program.

Few communities are fortunate enough to have sufficient revenues available at any given time to satisfy all demands for new or improved public facilities and services. Consequently, most are faced with the necessity of determining the relative priority of specific projects and establishing a program schedule for their initiation and completion. The orderly programming of public improvements is to be accomplished in conjunction with a long-range plan.

In essence, the Capital Improvements Program is simply a schedule for implementing public capital improvements, which acknowledges current and anticipated demands, and which recognizes present and potential financial resources available

to the community. The Capital Improvements Program is a major planning tool for assuring that public improvements proceed to completion in an efficient manner. The Capital Improvements Program is not intended to encourage the spending of additional public monies, but is simply a means by which an impartial evaluation of needs may be made. The Program typically is prepared for a five-year period, the first year of which is called Capital Improvements Budget.

Long-range programming of public improvements is based upon three fundamental considerations. First, the proposed projects must be selected on the basis of community need. Second, the program must be developed within the community's financial constraints and must be based upon a sound financial plan. Finally, program flexibility must be maintained through the annual review and approval of the capital budget. The strict observance of these conditions requires periodic analysis of various community development factors, as well as a thorough and continuing evaluation of all proposed improvements and related expenditures.

It is essential that in the process of preparing and developing the program, the Planning Commission be assigned a role in reviewing project proposals to assure conformity with the Master Plan and to make recommendations regarding prioritizing projects, and appropriate methods of financing.

9.2.5 Michigan Certified Community Recreation Plan

Any community can develop a community recreation plan. This plan helps to determine a community's future recreation needs, and proposes solutions to meet such needs. In addition to this, a certified community recreation plan allows the local government to apply for grant funds through the Michigan Department of Natural Resources (MDNR) such as the Michigan Natural Resources Trust Fund and Land and Water Conservation Fund.

9.2.6 Downtown Development Authority (DDA)

A local unit of government may create a Downtown Development Authority in order to halt property value deterioration, increase property tax valuation, and promote economic growth in a central business district. A DDA has the authority to create development plans, encourage historic preservation, authorize acquisition of property, and promote economic growth. A DDA may utilize Tax Increment Financing as a tool for generating revenue.

9.2.7 Special Assessment Districts

Special assessments are a fee levied by the community for the financing of local improvements that directly benefit the landowners that must pay the assessment. Taxes levied for public improvements within a Special Assessment District can be applied for such improvements as utilities, public roads, trash collection, bicycle paths, parks, sidewalks, lighting, and tree maintenance.

9.3 Grant and Loan Programs

Table 17 provides a listing of grant and loan programs organized by type of project. A detailed description of each program is included below.

Table 17
Grant Programs by Project Type

Road Improvements
1. Transportation Economic Development Fund
2. Public Works and Economic Development Program
3. Community Development Block Grant (CDBG)
Road Beautification
1. Transportation Enhancement Program (TEA-21)
Water and Sewer Systems
1. Public Works and Economic Development Program
2. Community Development Block Grant (CDBG)
3. Water and Wastewater Disposal Loan and Grant Program
Economic Development
1. Community Development Block Grant (CDBG)
2. Rural Business Enterprise Grants
Parks and Recreation
1. Michigan Natural Resources Trust Fund
2. Land and Water Conservation Fund
Non-Vehicular Trails
1. Michigan Natural Resources Trust Fund
2. Transportation Enhancement Program
Community Facilities
1. Community Facilities Program

9.3.1 Community Facilities Program

Agency

U.S. Department of Agriculture, Rural Housing Service

Description

This program is designed to provide loan guarantees, loans, and grants to construct, enlarge, or improve essential community facilities in rural areas. Examples of these facilities are fire and rescue facilities, jails, health clinics, nursing homes, airports, city halls, libraries, community centers, and schools.

Michigan Examples

Albee Township
Township Hall
Direct Loan, $500,000
Tawas City
Fire Hall
Direct Loan, $750,000

Deckerville
Deckerville Community Hospital, equipment
Guaranteed Loan, $350,000

Bay City
Health Delivery Inc, Medical Facility
Direct Loans, $1,390,000

Contact

Rural Development Field Office
1075 Cleaver Rd.
Caro, MI 48723
Phone: (989) 673-7588
Fax: (989) 673-1848

Eligibility

Municipalities, counties, special purpose districts, and nonprofit corporations.

Funding Available Nation Wide

FY 1999: $210,000,000 for loan guarantees
FY 1999: $169,475,983 for direct loans
FY 1999: $6,500,000 for grants

9.3.2 Public Works and Economic Development Program

Agency

Economic Development Administration, U.S. Department of Commerce

Description

This program empowers communities in economic decline to revitalize, expand, and upgrade their physical infrastructure to attract new industry, encourage business expansion, diversify local economies, and generate or retain long-term, private sector jobs and investment.

Michigan Examples

Kalkaska
Infrastructure development for industrial park
$312,000

Reed City
Water system improvements
$815,000

Boyne City
Water & sanitary sewer system for industrial park
$835,000

Farwell
Infrastructure development for industrial park
$665,000

Michigan Contact

State Office of the Economic Development Administration
Traverse City, MI
Phone: (231) 938-1712

Eligibility

Municipalities, economic development districts, higher learning institutions, and public and private nonprofit organizations.

Funding Available Nation Wide

FY 1999: $205,748,000
Average FY 1999 grant: $829,000

9.3.3 Rural Business Enterprise Grants

Agency

U.S. Department of Agriculture, Rural Business Services

Description

The Rural Business Enterprise Grants help to finance and facilitate the development of small and emerging private business enterprises. Costs that may be paid from grant funds include the acquisition of land, construction, utility extensions, professional services, and training fees, among others.

Michigan Project Examples

Iron County
Creation of a revolving loan fund for small businesses

Ontonagon County
Technical Assistance for businesses

Created for Caring, Bay City
Technical Training

Dowagiac
Water and Sewer Connections for small businesses

Contact

Rural Business & Cooperative Development Services Office
Lansing, MI
Phone: (517) 324-5220

Eligibility

Eligibility is limited to public bodies and private nonprofit corporations in rural areas with a population of less than 50,000.

9.3.4 Water and Wastewater Disposal Loan and Grant Program

Agency

U.S. Department of Agriculture

Description

This program provides loans, guaranteed loans, and grants for water, sewer, storm water, and solid waste disposal facilities in rural areas and towns of up to 10,000 people. Loan and grant funds may be used to: a) construct, repair, modify, expand, and improve water and waste facilities, b) acquire needed land, water sources, and water rights, and c) pay development costs for these facilities.

Michigan Examples

Pinconning
New sewer treatment plant

Elsie
Sewer system improvements

Bancroft
Water system improvements

Contact

Rural Development Field Office
1075 Cleaver Rd.
Caro, MI 48723
Phone: (989) 673-7588
Fax (989) 673-1848

Eligibility

All public entities – municipalities, counties, and special purpose districts in rural areas of up to 10,000 people.

Funding Available Statewide

FY 2000: Loan - $26,259,000
FY 2000: Grant - $13,008,000

9.3.5 Community Development Block Grant (CDBG)

Agency

Michigan Department of Commerce, Economic Development Corporation

Description

The CDBG program promotes community and economic revitalization through programs that support a wide variety of housing rehabilitation, public facilities and infrastructure, public services, and economic development projects, including brownfields redevelopment.

Michigan Examples

Hope Township, Midland County
Infrastructure improvements for business expansion project
$250,350

Village of Mayville, Tuscola County
Utilities and street improvements for manufacturing expansion project
$121,500

City of Lapeer
Infrastructure improvements for manufacturing company
$270,766

City of Yale, St. Clair County
Infrastructure improvements for industrial expansion project
$86,000

Contact

Business Finance Office, Economic Development Corporation
201 North Washington Square 3rd Floor
Lansing, MI 48913
Phone: (517) 335-1951

Eligibility

The non-entitlement community portion of the CDBG can apply to all local government units. The entitlement community portion of the CDBG applies to local government units with 50,000 or more residents.

Funding Available Nation Wide

FY 2000: $4.23 billion

9.3.6 Land and Water Conservation Fund (LWCF)

Agency

Michigan Department of Natural Resources

Description

Program provides grants to local units of government to acquire and develop land for outdoor recreation activities. Typical activities include land acquisition and park facilities development.

Contact

MDNR
Grants Administration Division
(517) 241-2480

Eligibility

Any unit of government including school districts with an approved MDNR Recreation Plan.

Funding Available

$15,000 grant minimum/$500,000 grant maximum for development projects

9.3.7 Michigan Natural Resources Trust Fund

Agency

Michigan Department of Natural Resources

Description

Provides grants for local units of government for the acquisition and development of lands and facilities for outdoor recreation or the protection of Michigan's significant natural resources.

Michigan Examples

Bad Axe
Buschlen Park Expansion
$136,500

Bay County
Boat Launch Expansion
$228,000

Meridian Township
Ferguson Park Extension
$103,875

Midland County
Pere Marquette Rail-Trail
$117,040

Contact

Regional Operations Office
Lansing, MI
Phone: (517) 241-4128

Eligibility

Any unit of government including school districts. Local units of government must have a DNR approved recreation plan.

Funding Available State Wide

FY 2000: $20-25 million
Individual grant: $15,000-$500,000 – no limits on land acquisition

9.3.8 Transportation Economic Development Fund

Agency

Michigan Department of Transportation

Description

The Transportation Economic Development Fund is a program that makes available funds for street improvements that are directly tied to economic expansion. Street construction within an industrial park is a common type of project funded with the help of the Transportation Economic Development Fund. There are six different project categories (A - F) within this fund, each relating to different project situations and municipality types.

Michigan Examples

Isabella County
Coldwater Rd. reconstruction
$300,000

Oscoda County
Abbe Rd. reconstruction
$146,000

Lapeer County
Graham Rd. reconstruction
$248,741

Contact

Office of Economic Development
P.O. Box 30050
Lansing, MI 48909
Phone: (517) 335-1069

Eligibility

The different project categories (A-F) have different eligibility requirements.

Funding Available Statewide

FY 1996: $23,424,351

9.3.9 Transportation Enhancement Program (TEA-21)

Agency

Michigan Department of Transportation

Description

Program is meant to promote transportation enhancement activities that go beyond traditional road improvements such as paving and widening. Typical projects include streetscaping, landscaping, signage, pedestrial walks, bicycle trails, scenic site acquisition, and restoration of historic transportation buildings.

Michigan Examples

Midland County
Pere Marquette Rail Trail
$420,000

Village of St. Charles
Construction and landscaping of median islands on East Belle Avenue
$24,000

Village of Homer
Creation of landscaped focal point in an old traffic circle
$10,720

Contact

MDOT
Enhancement Manager
(517) 335-2622

Eligibility

Grant must by sponsored by any government unit that receives fuel tax money.

9.4 Other Funding Options

9.4.1 Revenue Bonds

Revenue Bonds are those issued for various public improvement projects that are backed by the future revenues generated from the improvements. For example, water systems are improved by the sale of bonds, which are then repaid by water rates charged to the customers of the system. Sewer systems, airports, marinas, and other public facilities are often constructed by means of revenue bonds.

9.4.2 General Obligation Bonds

General Obligation Bonds are those that are issued by a municipality or other public body that are backed by the "full faith and credit" of that body. The municipality usually pledges its taxing ability, and therefore future tax revenues, to make the payments on the bonds. Because of the "full faith and credit" issue, and therefore a low amount of risk of default by the Township, these types of bonds usually attract the lowest interest rate.

9.4.3 Local Funds

In order to receive financial assistance from grant and loan programs, a community must also contribute a matching amount of funding for any project. It would be nearly impossible to receive funds from other sources without contributing some amount of general revenue funds as well. In general, the more a community contributes to a project, the more likely that community will receive grants and loans. Also, the more money a community contributes means that they might receive a greater amount of grant and loan funding.

9.4.4 Local Businesses

Seeking the help of your local businesses may result in much needed support for various public projects in the form of cash donations, material donations, and advertising, among others.

9.4.5 Local Foundations

Similar to the local businesses, local foundations might be able to provide support for civic projects.

9.5 Sources of Information

9.5.1 Michigan Townships Association

Provides training, newsletters, and consultations to Township officials.

512 Westshire Drive
P.O. Box 80078
Lansing, MI 48908
(517) 321-6467

9.5.2 County Extension Services

In Michigan, extension agents are concentrating on assisting communities with land use problems, free of charge. Agents may serve as meeting facilitators and may be able to access additional help from Michigan State University faculty.

Clare County Extension Services
County Building
P.O. Box 439
Harrison, MI 48625
(989) 539-7805

9.5.3 County Planning Commission

County Planning Commissions can be of help in coordinating regional and multi-jurisdictional planning efforts, and can be good sources of regional and local information.

Clare County Planning Commission
Clare County Courthouse
225 W. Main
Harrison, MI 48625
(989) 588-9343

9.5.4 Regional Planning Agency

Regional Planning Agencies were created in Michigan to act as coordinating agencies for the local governmental units they serve. These agencies create development plans, and conduct all types of research and studies for their planning region. Regional Planning Agencies also work to promote cooperation between the many different governmental units within their region. A local government may find their Regional Planning Agency to be a valuable resource for information and guidance. Municipalities in Roscommon County may seek guidance from:

East Central Michigan Planning and Development Region
3535 State St.
Saginaw, MI 48602
(989) 797-0800

9.5.5 Other Local Officials

Talking with other local units of governments with similar issues and problems can be extremely beneficial in determining successful land use policies and strategies.

9.5.6 Libraries

Your local library is always a good starting point as a source of information for a community. Historical records, historical business listings, and maps can all be utilized in local government research.

9.5.7 Internet

The Internet can be an invaluable resource for local communities. The following web sites may be useful:

American Planning Association – http://www.planning.org

Michigan Department of Environmental Quality – http://www.deq.state.mi.us

Michigan Department of Natural Resources – http://www.michigan.gov/dnr

Michigan Department of Transportation– http://www.michigan.gov/mdot

Michigan Economic Development Corporation – http://medc.michigan.org
Michigan Government Home Page – http://www.michigan.gov

Michigan Townships Association – http://www.mta-townships.org

Planning and Zoning Center – http://www.pzcenter.com

U.S. Department of Housing and Urban Development – http://www.hud.gov

U.S. Census Bureau – http://www.census.gov

U.S. Environmental Protection Agency – http://www.epa.gov

U.S. Department of Agriculture – http://www.usda.gov